

Compute-Adaptive Risk Control: A Finite-Sample Calibration Layer for Early-Exit and Cascade Inference

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ABSTRACT Adaptive-inference systems—early-exit networks and model cascades—reduce average computation by terminating easy inputs before the most expensive model path. Their exit thresholds are commonly selected by validation-set inspection, but the selected policy is data-dependent: the same calibration sample screens the candidates and chooses the deployed threshold, so empirical acceptability does not imply finite-sample control of deployment risk. We study *compute-adaptive risk control* (CARC), a finite-family calibration layer for a fixed adaptive model. Given a risk target α , an error probability δ , and a compute-ordered candidate family, CARC returns a certified threshold policy satisfying $R(\tau) \leq \alpha$ with probability at least $1 - \delta$ over the calibration draw, or reports that no candidate certifies. The statistical reduction to hypothesis testing is inherited from Learn-then-Test; the additional contributions are (i) a finite-sample cost characterization for fixed-sequence certification on compute chains, (ii) a multi-chain selector that searches several pre-specified per-exit chains with chain-level error allocation, (iii) a joint risk—compute certificate for pre-committed budgets, (iv) an estimated-weight sensitivity certificate for covariate shift, and (v) explicit deployment semantics that separate risk validity from target feasibility. The evidence is scoped to the implemented caches and simulations. In controlled simulation with oracle risks, certified selection stays below the tested violation levels while uncorrected validation tuning violates at 0.231–0.351. In a six-chain synthetic experiment, multi-chain CARC keeps true violation at 0–0.001 and lowers cost relative to Holm and Bonferroni in every tested cell. Real-cache studies on digits, breast-cancer, CIFAR-10, and CIFAR-100 show that richer pre-specified chains can reduce certified compute when feasible policies exist, while the same procedure reports infeasibility when the trained model cannot meet the requested target.

INDEX TERMS Adaptive inference, calibration, early-exit networks, finite-sample guarantees, risk control, statistical learning.

I. INTRODUCTION

Adaptive-depth inference—early-exit networks, cascades, and routers—reduces average compute by allowing inputs to terminate before the most expensive model path. The standard deployment workflow trains the adaptive model, evaluates candidate thresholds on held-out data, and deploys the cheapest threshold whose empirical risk appears acceptable. This workflow is statistically incomplete. Threshold selection is a post-selection operation: the calibration sample is used both to screen candidate policies and to select the deployed policy. Consequently, a threshold with empirical risk below α can have true risk above α . The same workflow also lacks a formal outcome for target infeasibility; when no candidate is safe

for the trained system and requested risk level, returning an uncertified fallback obscures the failure mode.

We study the following finite-sample problem: **given a fixed adaptive inference system, a finite calibration set, a risk target α , and an error probability δ , select a low-compute threshold policy while controlling the probability of deploying an unsafe policy.** CARC solves this problem by instantiating Learn-then-Test (LTT) for finite, compute-ordered threshold families [1]. The LTT reduction supplies valid p-values and family-wise error control; the adaptive-inference structure supplies the candidate order, the compute objective, and the need for an explicit infeasibility report. Relative to uncorrected validation tuning, CARC replaces

empirical acceptability with a finite-sample post-selection risk certificate. Relative to other valid selectors, CARC does not promise order-of-magnitude savings; its contribution is to align the testing order and deployment objective, then quantify the remaining efficiency cost.

The empirical objective is therefore calibration-specific rather than model-accuracy-specific. We do not claim state-of-the-art backbone accuracy. We test whether certified selection controls unsafe-output frequency, measure the compute premium of certification relative to uncorrected validation tuning, and expose infeasibility when the trained model cannot meet the requested target. In the fixed-grid digits real-cache study, the final exit is below the target grid before calibration and the certified chain stays within the tested violation levels; at $\alpha = 0.10$, naive validation inspection violates at 0.486–0.571. A CIFAR-10 branchy cache provides a feasible neural-cache check: for $n_{\text{cal}} = 2000$ and $\alpha = 0.15$ –0.30, multi-chain CARC selects in 0.956–1.000 of trials, keeps held-out violation at 0.001–0.017, and is cheaper than scalar-chain CARC in all tested cells. The CIFAR-100 branchy-cache run is a feasibility diagnostic: the pre-defined targets are mostly infeasible because the minimum full-pool chain risk is 0.3055, and the feasible-target cells are exploratory.

Scope of evidence. The guarantees in Sections 3–6 are proved in Appendix A. Controlled simulation with oracle risks verifies the $1 - \delta$ behavior and isolates the failure mode of naive post-selection tuning. A multi-chain simulation then tests whether a richer pre-specified family improves certified efficiency without losing validity. Real-cache studies on scikit-learn digits and breast-cancer data exercise the cache evaluator on feasible target grids; these staged tabular cascades validate the calibration layer, not deep-network accuracy. CIFAR-10 supplies a feasible neural early-exit cache. CIFAR-100 tests the opposite regime: a weaker branchy model whose requested targets mostly lie below the finite-pool risk frontier. Larger ImageNet-scale early-exit networks, language cascades, measured latency, and non-synthetic shift remain outside the evidence base of this submission.

Figure 1 summarizes the deployment logic before the formal notation. The model is fixed before calibration; CARC uses the calibration cache only to certify a pre-specified chain, while the naive branch inspects the same cache without a family-wise risk certificate.

A. CONTRIBUTIONS

The reduction "risk control = multiple testing" and the admissibility of fixed-sequence and graphical tests are due to LTT and the multiple-testing literature; Table 1 separates inherited machinery from the new compute-adaptive specialization. The contribution of this paper is a calibration-and-deployment layer for compute-adaptive inference, with the following components.

1) Finite-family formulation for adaptive inference.

We cast threshold selection for early-exit and cascade policies as risk control over a finite, compute-ordered candidate family. The formulation applies to

CARC flow

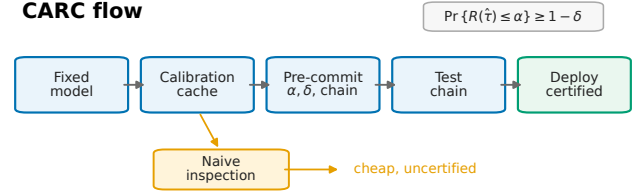


FIGURE 1. CARC calibration/deployment flow. A fixed multi-exit model produces cached calibration scores, losses, and compute values. The user pre-commits (α, δ) and a compute-ordered chain; LTT-style tests certify configurations from expensive to cheap; the deployed policy is the cheapest certified one, or the procedure reports infeasibility. The orange branch shows the naive validation-inspection baseline: it may be cheap, but it does not provide the family-wise certificate used for risk control.

any bounded loss $\ell_{\text{loss}} \in [0, 1]$, including 0–1 error, selective risk, and class-conditional error rates.

- 2) **Cost characterization for fixed-sequence certification.** The validity guarantee is inherited from LTT and carries no chain-length penalty. The new statement in Theorem 2(b) is an efficiency characterization: under a monotone-risk condition, the selected policy lies within a finite-sample boundary band of the cheapest safe policy in the chosen chain, with width $\Delta_n = O(\sqrt{\ln(K/\delta)/n})$.
- 3) **Multi-chain selector for richer policy families.** We extend the one-dimensional scalar-threshold chain to several pre-specified per-exit threshold chains. The procedure runs fixed-sequence testing within each chain at an allocated level, unions the certified sets, and controls FWER by a chain-level union bound. This searches more policies than the scalar chain while avoiding the full δ/K penalty of Bonferroni when the number of chains is much smaller than the number of configurations.
- 4) **Joint risk-compute certificate.** Theorem 3 certifies $R(\tau) \leq \alpha$ and $C(\tau) \leq B$ for settings where a compute budget must be fixed before deployment. The risk certificate remains the primary guarantee; the compute certificate is a conservative pre-commitment device rather than a substitute for measuring realized latency.
- 5) **Estimated-weight shift certificate.** Theorem 4 gives a finite-sample sufficient condition under covariate shift when importance weights are estimated on a separate fold and an L^1 error budget η is supplied. The result is a sensitivity certificate: validity depends on the correctness of the budget, and feasibility can degrade as η increases.
- 6) **Certify-or-report-infeasibility deployment discipline.** We separate three quantities that validation tuning conflates—risk validity, target feasibility for the trained system, and compute efficiency—and require *infeasibility* rather than an uncertified fallback when no candidate certifies. The CIFAR-100 study documents this behavior under a weak branchy model: CARC certifies no policy when no pre-defined target admits

TABLE 1. What CARC borrows from Learn-then-Test (LTT) and the multiple-testing literature, and what it adds. The validity machinery is inherited; the new content is the compute-chain specialization and the certificates and deployment semantics built on it.

Ingredient	Source	Role in CARC
Risk control = multiple testing	Borrowed	Foundation
Concentration p-values (Hoeffding, HB, emp. Bernstein)	Borrowed	Per-config tests
FWER rules (Bonferroni, fixed-sequence, Holm)	Borrowed	Certification
Compute-ordered chain from exit structure	New	Supplies test order
Several pre-specified chains	New	Richer selector
Two-sided cost bound (Th. 2(b))	New	Efficiency band
Joint risk–compute certificate (Th. 3)	New	Pre-committed budget
Estimated-weight shift certificate (Th. 4)	New	Shift sensitivity
Certify-or-report-infeasibility	New	Deployment semantics

a safe configuration, whereas uncorrected validation tuning selects unsafe policies in 28–46% of trials and violates the target in exactly those trials.

- 7) **Implementation and validation.** The reference implementation covers the p-values, certification procedures, selectors, threshold-family builders, and cache evaluators. Experiments verify the guarantee in simulation with oracle risks, evaluate the multi-chain redesign in synthetic and real-cache settings, confirm feasible-cache behavior on tabular cascades and a CIFAR-10 branchy cache, and document infeasibility reporting on a CIFAR-100 branchy cache whose pre-defined targets are mostly unattainable.

B. RELATION TO LEARN-THEN-TEST

Borrowed. From LTT [1] and the multiple-testing literature [2]–[4], we use the reduction from risk control to testing nulls $H_\tau : R(\tau) > \alpha$, valid concentration p-values, and family-wise error rate (FWER) procedures for finite hypothesis families. The Hoeffding–Bentkus and empirical-Bernstein p-values are also standard ingredients [5]–[7].

New. We specialize this machinery to adaptive inference and analyze the effect of a compute-ordered candidate family. Theorem 2(b) characterizes the efficiency loss of fixed-sequence certification on an exit/cascade chain; the multi-chain selector below searches several pre-specified per-exit chains with a chain-level error split; Theorem 3 adds a conservative joint risk–compute certificate for pre-committed budgets; Theorem 4 gives an estimated-weight sufficient condition for covariate shift; and the implementation evaluates these claims with controlled simulation, multi-chain redesign experiments, feasible real-cache confirmations, and CIFAR branchy-cache diagnostics. The contribution is a calibrated threshold-selection methodology, not a new p-value. When the chain order is doubtful, we implement Holm’s step-down procedure [2]; the graphical family in [4] is a broader unimplemented option.

II. PROBLEM SETUP AND NOTATION

Let $(X, Y) \sim P$ with $X \in \mathcal{X}$, $Y \in \mathcal{Y}$. A multi-exit network has exits $\ell \in \{1, \dots, L\}$ ordered by depth. Exit ℓ produces a prediction $\hat{y}_\ell(x)$ and a scalar confidence $s_\ell(x) \in [0, 1]$ (e.g., top softmax probability, or $1 - \text{entropy}$). Let $c_\ell \geq 0$ be the cumulative compute (FLOPs, or wall-clock proxy) to *reach and evaluate* exit ℓ , with $c_1 < c_2 < \dots < c_L =: C_{\max}$.

Exit policy. A configuration is a threshold vector $\tau = (\tau_1, \dots, \tau_{L-1}) \in [0, 1]^{L-1}$. On input x , the policy exits at

$$E_\tau(x) = \min \left(\{ \ell < L : s_\ell(x) \geq \tau_\ell \} \cup \{L\} \right),$$

i.e., it stops at the first exit whose confidence clears its threshold, and otherwise runs to the final exit L . The deployed prediction is $\hat{Y}_\tau(x) = \hat{y}_{E_\tau(x)}(x)$.

Risk and compute functionals. For a bounded loss $\ell_{\text{loss}} : \mathcal{Y} \times \mathcal{Y} \rightarrow [0, 1]$,

$$R(\tau) = \mathbb{E}_P[\ell_{\text{loss}}(\hat{Y}_\tau(X), Y)], \quad C(\tau) = \mathbb{E}_P[c_{E_\tau(X)}].$$

Examples of ℓ_{loss} : 0–1 error $\mathbf{1}\{\hat{Y}_\tau \neq Y\}$; selective risk on accepted points; class-conditional false-negative rate (clip to $[0, 1]$). All that the theory needs is boundedness.

Calibration data. We observe n i.i.d. draws $\mathcal{D} = \{(X_i, Y_i)\}_{i=1}^n \sim P$, disjoint from training. Define empirical estimates

$$\begin{aligned} \hat{R}(\tau) &= \frac{1}{n} \sum_{i=1}^n \ell_{\text{loss}}(\hat{Y}_\tau(X_i), Y_i), \\ \hat{C}(\tau) &= \frac{1}{n} \sum_{i=1}^n c_{E_\tau(X_i)}. \end{aligned}$$

Goal. Given target risk $\alpha \in (0, 1)$ and confidence level $\delta \in (0, 1)$, return $\hat{\tau}$ (a measurable function of $\mathcal{D}$) such that

$$\Pr_{\mathcal{D}}(R(\hat{\tau}) \leq \alpha) \geq 1 - \delta, \quad (\text{P1})$$

and, among configurations satisfying (P1), $\hat{\tau}$ has small compute $C(\hat{\tau})$. A stronger *dual* specification adds a budget B :

$$\Pr_{\mathcal{D}}(R(\hat{\tau}) \leq \alpha \text{ and } C(\hat{\tau}) \leq B) \geq 1 - \delta. \quad (\text{P2})$$

Grid. We work over a finite candidate set $\Lambda \subset [0, 1]^{L-1}$ (e.g., a per-exit grid of m thresholds, $|\Lambda| \leq m^{L-1}$, or a scalar-parameterized chain; see Section 4). All probabilities below are over the draw of $\mathcal{D}$; $\hat{\tau}$ is data-dependent.

Nature of the guarantee. (P1) is a *training-conditional* (PAC-style) statement: with probability $\geq 1 - \delta$ over the calibration draw, the *fixed, deployed* policy $\hat{\tau}$ has true risk $\leq \alpha$ for all subsequent inputs. It is not a marginal-over-future-test-points guarantee that averages away calibration noise; it is the stronger object an operator wants, since $\hat{\tau}$ is frozen at deployment. The qualifier *distribution-free* applies to **validity**: it holds for any P . It does **not** extend to the efficiency (cost-optimality) and shift results, which require,

respectively, a monotonicity condition (Section 4.1) and a weight-error budget (Section 6.2); we flag this wherever it matters.

III. RISK CONTROL AS MULTIPLE TESTING

We adapt Learn-then-Test (LTT) [1]. For each $\tau \in \Lambda$ define the null

$$H_\tau : R(\tau) > \alpha \quad (\text{"}\tau \text{ is unsafe"}).$$

Rejecting H_τ certifies τ as safe. We need (a) a valid p-value p_τ for each H_τ , and (b) a family-wise-error-controlling (FWER) procedure that maps $\{p_\tau\}$ to a certified set $\hat{\Lambda} \subseteq \Lambda$ with

$$\Pr(\exists \tau \in \hat{\Lambda} : R(\tau) > \alpha) \leq \delta. \quad (1)$$

Given (1), any selection rule restricted to $\hat{\Lambda}$ inherits validity, because the bad event "we output an unsafe τ " is contained in " $\hat{\Lambda}$ contains an unsafe τ ."

A. VALID P-VALUES FROM CONCENTRATION

Because $\ell_{\text{loss}} \in [0, 1]$ and the n losses are i.i.d. given τ , $\hat{R}(\tau)$ is a mean of bounded i.i.d. variables. The one-sided Hoeffding inequality [5] gives, for the boundary mean $R(\tau) = \alpha$ and any observed $\hat{R}(\tau) \leq \alpha$,

$$\Pr_{R(\tau)=\alpha}(\hat{R}(\tau) \leq \hat{r}) \leq \exp(-2n(\alpha - \hat{r})^2).$$

Since under H_τ we have $R(\tau) \geq \alpha$ (taking the closure), the supremum of the left side over the null is attained at $R(\tau) = \alpha$. Hence

$$p_\tau = \exp(-2n(\alpha - \hat{R}(\tau))_+^2) \quad (2)$$

is a valid (super-uniform under H_τ) p-value, where $(z)_+ = \max(z, 0)$. Tighter choices—the **Hoeffding–Bentkus (HB)** p-value used in LTT [1], [6], or an empirical-Bernstein p-value [7] when the loss has small variance—slot in directly and improve power; we use (2) for exposition and HB in experiments.

B. BONFERRONI LTT (ASSUMPTION-FREE BASELINE)

Reject H_τ iff $p_\tau \leq \delta/|\Lambda|$, giving $\hat{\Lambda}_{\text{Bonf}} = \{\tau : p_\tau \leq \delta/|\Lambda|\}$. Then output

$$\hat{\tau} = \arg \min_{\tau \in \hat{\Lambda}_{\text{Bonf}}} \hat{C}(\tau) \quad \text{if } \hat{\Lambda}_{\text{Bonf}} \neq \emptyset,$$

and otherwise report infeasibility for the requested $(\alpha, \delta, n, \Lambda)$ without issuing a certified deployment policy.

Theorem 1 (Distribution-free risk validity). *For losses in $[0, 1]$, i.i.d. calibration data, p-values (2), and the Bonferroni rule at level δ , the probability that the procedure certifies and outputs an unsafe configuration is at most δ : $\Pr(\text{output exists and } R(\hat{\tau}) > \alpha) \leq \delta$. If the certified set is empty, it reports infeasibility rather than claiming a risk guarantee.*

Proof. By Bonferroni, $\Pr(\exists \tau \in \hat{\Lambda}_{\text{Bonf}} : R(\tau) > \alpha) \leq \sum_{\tau: R(\tau) > \alpha} \Pr(p_\tau \leq \delta/|\Lambda|) \leq |\{\tau : H_\tau\}| \cdot \delta/|\Lambda| \leq \delta$, using super-uniformity of p_τ under H_τ . On the complementary event every certified τ is safe; if the procedure outputs, $\hat{\tau} \in \hat{\Lambda}_{\text{Bonf}}$, so $R(\hat{\tau}) \leq \alpha$. If $\hat{\Lambda}_{\text{Bonf}} = \emptyset$, the procedure reports infeasibility and no certified-risk statement is attached to any operational fallback unless that fallback is separately certified. $\square$

The defect of Bonferroni is the $\delta/|\Lambda|$ penalty: with a fine per-exit grid $|\Lambda|$ is large, the threshold is tiny, and the certified set is small and expensive. The next section avoids this penalty on the *validity* side by testing a pre-ordered chain — at the price of restricting the search to that chain, a trade-off we make explicit.

IV. THE COMPUTE-CHAIN INSTANTIATION

Fixed-sequence testing is a standard FWER procedure, already named in LTT for monotone, pre-orderable hypotheses [1], [3]; this section does **not** propose it as new. What it contributes is that early-exit deployment supplies the pre-specified order without extra design, aligned with the deployment objective, plus a two-sided cost bound (Theorem 2(b)) on how close the resulting policy is to the cheapest safe policy in the chain.

A. THE MONOTONE STRUCTURE OF EXIT POLICIES

Order configurations by **compute**. Consider a *chain* of configurations $\tau^{(1)} \preceq \tau^{(2)} \preceq \dots \preceq \tau^{(K)}$ that is increasing in compute, i.e. $C(\tau^{(1)}) \leq \dots \leq C(\tau^{(K)})$, with $\tau^{(K)} = 1$ the always-run-to- L policy. A natural and common construction is a **scalar-parameterized** family $\tau(t) = (t, \dots, t)$ for t on a grid $0 = t_1 < \dots < t_K = 1$: raising the global threshold t makes every exit more conservative, pushing more inputs to deeper exits, which *increases* compute monotonically (deterministically, since each input's exit index is nondecreasing in t). More refined chains (per-exit thresholds raised in a fixed schedule) are also admissible.

Assumption M (monotone risk along the chain). $R(\tau^{(1)}) \geq R(\tau^{(2)}) \geq \dots \geq R(\tau^{(K)})$; equivalently, deeper-on-average policies have weakly lower risk.

Assumption M says deeper exits are, on average, at least as accurate — the premise that motivates deploying a deep model at all. Its empirical version (monotone $\hat{R}$ along the chain) is observable, and we report it; the assumption itself concerns the true R . Crucially, Assumption M is needed only for the *efficiency* statement of Theorem 2(b), never for validity.

Within-chain scope. A chain is a one-dimensional curve through the $(L-1)$ -dimensional configuration space, and the scalar family $\tau(t) = (t, \dots, t)$ in particular is restrictive: per-exit thresholds can dominate a single shared threshold. Consequently "cheapest safe policy" in Theorem 2 means cheapest *along the chosen chain*, not the global optimum over all of $[0, 1]^{L-1}$. The chain is a deliberate design choice trading search breadth against the multiplicity penalty: a Holm step-down over the full grid recovers more configurations at

a $\log|\Lambda|$ cost (Section 4.2, remark), and the more general graphical family is a possible extension. We report the chain and Holm variants and let the practitioner pick between those implemented procedures.

B. CERTIFICATION ALONG THE CHAIN

Apply LTT's fixed-sequence test to the compute chain, walking from most expensive to cheapest, $\tau^{(K)}, \tau^{(K-1)}, \dots$, and **stopping at the first configuration that fails to certify**. With the Hoeffding p-value (2), the certification test $p_{\tau^{(k)}} \leq \delta$ is exactly

$$\hat{R}(\tau^{(k)}) \leq \alpha - \epsilon_n(\delta), \quad \epsilon_n(\delta) := \sqrt{\frac{\ln(1/\delta)}{2n}}.$$

Let k^* be the cheapest index reached before stopping; output $\hat{\tau} = \tau^{(k^*)}$ if at least one configuration certifies. If even $\tau^{(K)}$ fails to certify, report the task infeasible at this (α, δ, n) and do not claim a certified deployment policy. Running $\tau^{(K)} = \mathbf{1}$ can be a conservative engineering fallback, but it is not a risk guarantee unless $\tau^{(K)}$ itself certifies or is separately known to satisfy $R(\tau^{(K)}) \leq \alpha$.

Define the cheapest truly-safe index $k_0 := \min\{k : R(\tau^{(k)}) \leq \alpha\}$.

Theorem 2 (validity and a two-sided cost bound).

(a) *Validity (LTT/fixed-sequence; stated for completeness).* For p-values valid under each $H_{\tau^{(k)}}$ and the fixed-sequence rule at level δ on the pre-specified chain order, the probability of certifying and outputting an unsafe configuration is at most δ , with no dependence on K .

(b) *Cost bound (our contribution).* On the validity event the output never undercuts the safe frontier: $C(\hat{\tau}) \geq \min_{k: R(\tau^{(k)}) \leq \alpha} C(\tau^{(k)})$, and under Assumption M this means $k^* \geq k_0$. For the other direction, let

$$\Delta_n := \epsilon_n(\delta) + \epsilon_n(\delta/K) = O\left(\sqrt{\frac{\ln(K/\delta)}{n}}\right),$$

$$k_1 := \min\{k : R(\tau^{(j)}) \leq \alpha - \Delta_n \text{ for all } j \geq k\}.$$

Then $\Pr(k^* \leq k_1) \geq 1 - \delta$. Combining, on an event of probability $\geq 1 - 2\delta$,

$$k_0 \leq k^* \leq k_1,$$

so the excess compute of the certified policy over the cheapest safe policy in the chain is at most $C(\tau^{(k_1)}) - C(\tau^{(k_0)})$ — the compute spanned by configurations whose true risk lies in the boundary band $(\alpha - \Delta_n, \alpha]$. As $n \rightarrow \infty$, $\Delta_n \rightarrow 0$ and $k_1 \rightarrow k_0$.

Proof. (a) is the standard fixed-sequence guarantee: since the order is fixed in advance and we stop at the first non-rejection, an error (certifying and outputting an unsafe config) requires rejecting the first true null in the walk, which has probability $\leq \delta$ by super-uniformity; this is independent of K .

(b) *Lower bound.* $\hat{\tau}$ is certified, hence safe on the validity event, so its compute is at least that of the cheapest safe

configuration; under Assumption M the safe set is the suffix $\{k \geq k_0\}$ and compute is increasing in k , giving $k^* \geq k_0$. *Upper bound.* The walk reaches index k_1 iff every config in $\{k_1, \dots, K\}$ certifies, i.e. $\hat{R}(\tau^{(j)}) \leq \alpha - \epsilon_n(\delta)$ for all $j \geq k_1$. For any such j , $R(\tau^{(j)}) \leq \alpha - \Delta_n = \alpha - \epsilon_n(\delta) - \epsilon_n(\delta/K)$, so

$$\begin{aligned} \Pr(\hat{R}(\tau^{(j)}) > \alpha - \epsilon_n(\delta)) \\ &\leq \Pr(\hat{R}(\tau^{(j)}) - R(\tau^{(j)}) > \epsilon_n(\delta/K)) \\ &\leq e^{-2n \epsilon_n(\delta/K)^2} = \delta/K \end{aligned}$$

by the upper-tail Hoeffding inequality. A union bound over the at most K such indices gives $\Pr(\exists j \geq k_1 : \hat{R}(\tau^{(j)}) > \alpha - \epsilon_n(\delta)) \leq \delta$, i.e. $\Pr(k^* \leq k_1) \geq 1 - \delta$. Intersecting with the validity event and a union bound yields the sandwich with probability $\geq 1 - 2\delta$. $\square$

The structure of the bound is the main theoretical distinction: **validity carries no K -penalty** (part a), while the **cost of searching a length- K chain appears as an $O(\sqrt{\ln K/n})$ efficiency band** (part b). Early stopping is therefore a validity device, not an exact optimizer. A safe configuration within Δ_n of the boundary may fail to certify and halt the walk, returning a more expensive certified policy, but not an unsafe one. Relative to Bonferroni over the same grid, the chain uses per-test level δ rather than δ/K , which tightens $\epsilon_n(\delta)$ and shrinks the band; this advantage follows from LTT once the adaptive-inference system supplies a fixed compute order.

Magnitude of the chain-vs-Bonferroni gain. The advantage enters through the *difference* between $\epsilon_n(\delta)$ and $\epsilon_n(\delta/K)$, i.e. $\sqrt{\ln(1/\delta)/2n}$ vs. $\sqrt{\ln(K/\delta)/2n}$. This difference is modest unless K is large or n is small; in the controlled simulation of Section 10.1, the certified-compute gap between chain and Bonferroni is a few percent. The bound therefore supports a calibrated interpretation: fixed-sequence testing removes the chain-length penalty from validity and can reduce the efficiency band relative to Bonferroni, but the realized compute premium over an invalid naive selector is empirical and must be reported for each model, target, and chain. The chain provides zero chain-length penalty on *validity* and an expensive-to-cheap testing order aligned with deployment. It does *not* imply large compute savings over other *valid* selectors such as Holm; the large compute gaps reported in Section 10 are against the *invalid* naive selector.

Figure 2 visualizes the two-sided cost statement. The important point is not exact recovery of the cheapest safe index k_0 , but the separation between validity and efficiency: validity prevents moving left of the safe frontier, while the finite-sample search price appears as a boundary band in compute.

Remark (assumption-free fallback when Assumption M is doubtful).

When per-exit thresholds are tuned jointly, or the empirical-monotonicity diagnostic looks suspect, drop the chain for an FWER procedure that needs no ordering. We implement and evaluate **Holm's step-down** [2], which is valid unconditionally, requires no monotonicity, and strictly dominates Bonferroni. The general *graphical* family in [4] subsumes both the chain

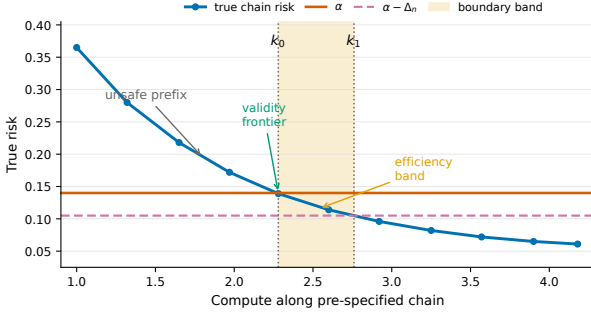


FIGURE 2. Schematic of the compute-chain cost bound. This is an illustrative monotone-risk chain, not an experimental curve. The fixed-sequence test walks from expensive to cheap. On the validity event, the selected configuration does not undercut the cheapest safe in-chain policy k_0 . Under Assumption M, the additional compute is bounded by the span between k_0 and k_1 , where k_1 is determined by the boundary band $(\alpha - \delta_n, \alpha]$.

and Holm and can recycle significance along the compute partial order—reducing to the chain test under exact monotonicity at a $\log |\Lambda|$ band—but we do not implement or evaluate it here, and do not claim its specific behavior.

C. MULTI-CHAIN CARC

The scalar chain $\tau(t) = (t, \dots, t)$ is simple but restrictive: avoiding a weak early exit may require a high threshold there while keeping later, more accurate exits permissive. Full-grid Bonferroni or Holm can search these per-exit combinations, but they pay a multiplicity price over every configuration. Multi-chain CARC uses an intermediate design. Before seeing calibration risks, specify M compute-ordered chains

$$\Lambda_m = \{\tau_m^{(1)} \preceq \dots \preceq \tau_m^{(K_m)}\}, \quad m = 1, \dots, M,$$

for example several per-exit threshold schedules. Let non-negative weights w_m satisfy $\sum_m w_m \leq 1$. Run fixed-sequence testing from expensive to cheap within chain m at level $\delta_m = w_m \delta$, union the certified sets

$$\hat{\Lambda}_{MC} = \bigcup_{m=1}^M \hat{\Lambda}_m,$$

and output the lowest-cost configuration in $\hat{\Lambda}_{MC}$; if the union is empty, report infeasibility. Uniform weights give $\delta_m = \delta/M$. The procedure reduces to the scalar chain when $M = 1$ and to a Bonferroni-like rule when every chain has one configuration, but in the common case $M \ll \sum_m K_m$ it spends error probability across chains rather than across every candidate.

Proposition 1 (multi-chain validity). Assume every chain order is fixed before calibration and each per-configuration p -value is valid for $H_\tau : R(\tau) > \alpha$. Multi-chain CARC controls the probability of outputting any unsafe certified configuration at level δ :

$$\Pr(\exists \tau \in \hat{\Lambda}_{MC} : R(\tau) > \alpha) \leq \delta.$$

Proof. For chain m , fixed-sequence testing at level δ_m controls the probability of certifying any unsafe configuration within that chain by δ_m , exactly as in Theorem 2(a). Therefore

$$\begin{aligned} & \Pr(\exists \tau \in \hat{\Lambda}_{MC} : R(\tau) > \alpha) \\ & \leq \sum_{m=1}^M \Pr(\exists \tau \in \hat{\Lambda}_m : R(\tau) > \alpha) \\ & \leq \sum_{m=1}^M \delta_m \leq \delta, \end{aligned}$$

where the first inequality is Boole's inequality and no independence across chains is required. Any downstream selection restricted to the union inherits the same validity. $\square$

Reading the guarantee. Multi-chain CARC is not an adaptive search over arbitrary schedules. The chains, their order, and the weights must be fixed before calibration risks are inspected. The method covers several engineering-relevant threshold schedules while paying a chain-count penalty rather than a configuration-count penalty. If the pre-specified chains miss the globally cheapest safe policy, validity remains intact but efficiency is bounded only relative to the searched family.

V. JOINT RISK-COMPUTE CONTROL

Problem (P2) additionally certifies a compute budget B . The motivation is pre-commitment: an operator who must provision hardware, cap energy, or sign a latency service-level agreement may need a budget certificate before deployment, whereas realized compute can be measured after deployment. This certificate is secondary to the risk guarantee and is conservative under the range-based bound used here. Compute $c_{E_\tau(X)} \in [0, C_{\max}]$ is bounded, so $\hat{C}(\tau)$ concentrates by Hoeffding: for a single τ , with probability $\geq 1 - \delta_C$,

$$C(\tau) \leq \hat{C}(\tau) + C_{\max} \sqrt{\frac{\ln(1/\delta_C)}{2n}} =: U_C(\tau; \delta_C).$$

To hold uniformly over the data-dependent choice, take a union bound over the K chain points (or $|\Lambda|$ grid points), replacing δ_C by the deterministic δ_C/K (resp. $\delta_C/|\Lambda|$).

Algorithm (dual control). Split $\delta = \delta_R + \delta_C$. (i) Certify risk: form $\hat{\Lambda}$ as in Section 3–Section 4 at level δ_R . (ii) Restrict to budget-feasible configs using the uniform compute bound: $\hat{\Lambda}_B = \{\tau \in \hat{\Lambda} : U_C(\tau; \delta_C/K) \leq B\}$. (iii) Output $\hat{\tau} = \arg \min_{\tau \in \hat{\Lambda}_B} \hat{C}(\tau)$, or report infeasible if $\hat{\Lambda}_B = \emptyset$.

Theorem 3 (Joint guarantee). With the split budget $\delta = \delta_R + \delta_C$ and the procedure above,

$$\Pr(R(\hat{\tau}) \leq \alpha \text{ and } C(\hat{\tau}) \leq B) \geq 1 - \delta.$$

Proof. Let $\mathcal{A} = \{\forall \tau \in \hat{\Lambda} : R(\tau) \leq \alpha\}$ and $\mathcal{B} = \{\forall \tau \in \hat{\Lambda} : C(\tau) \leq U_C(\tau; \delta_C/K)\}$ (the compute event ranges over the fixed grid, with the deterministic correction δ_C/K). By Section 3–Section 4, $\Pr(\mathcal{A}^c) \leq \delta_R$; by Hoeffding and a union bound over the K points, $\Pr(\mathcal{B}^c) \leq \delta_C$. On $\mathcal{A} \cap \mathcal{B}$: $\hat{\tau} \in \hat{\Lambda} \Rightarrow R(\hat{\tau}) \leq \alpha$, and $\hat{\tau} \in \hat{\Lambda}_B \Rightarrow C(\hat{\tau}) \leq U_C(\hat{\tau}; \delta_C/K) \leq B$. Union bound: $\Pr((\mathcal{A} \cap \mathcal{B})^c) \leq \delta_R + \delta_C = \delta$. $\square$

A symmetric formulation — *minimize risk subject to a hard budget* — swaps the roles, certifying a budget by inverting the compute bound and minimizing $\hat{R}$ over budget-feasible configs; both are special cases of certifying one functional and optimizing the other.

On the split (δ_R, δ_C) . We do not present the split as a contribution. Because δ_R, δ_C enter only through $\sqrt{\ln(1/\cdot)}$, moving error probability between risk and compute changes each slack only logarithmically. We default to $\delta_R = 0.9\delta$, $\delta_C = 0.1\delta$, allocating most power to the risk certificate, and verify in E5 that performance is flat across a broad central range. If a deployment requires a tuned split, the choice reduces to a one-dimensional search.

Looseness of the compute bound. The range- $C_{\max}$ Hoeffding UCB U_C is conservative, since per-input cost is far from uniform on $[0, C_{\max}]$; in our simulation the dual certificate is reported infeasible whenever the budget B is set near the realized compute, even when a budget-respecting safe policy exists. An empirical-Bernstein compute bound (variance-adaptive, as in Section 6.1) tightens U_C materially and is the recommended form in practice; we keep Hoeffding in the statement for transparency. This is a further reason we treat the compute certificate as secondary to the risk certificate.

VI. COVARIATE-SHIFT ROBUSTNESS

This section develops a formal sensitivity-analysis framework for covariate shift. It identifies the boundary under which shift-robust certification remains valid when importance weights are estimated: sample splitting prevents estimator reuse, an externally justified L^1 weight-error budget controls bias, and increasing that budget can make certification infeasible.

Suppose deployment is under Q with $X \sim Q_X$, $Y \mid X$ unchanged, and likelihood ratio $w(x) = dQ_X/dP_X$. The target is $R_Q(\tau) = \mathbb{E}_Q[\ell_{\text{loss}}] = \mathbb{E}_P[w(X)\ell_{\text{loss}}]$. The exact-weight case (Section 6.1) is standard; Section 6.2 handles the realistic case where w is estimated.

A. EXACT WEIGHTS (WARM-UP)

If $0 \leq w \leq W$ is known exactly, then $w\ell_{\text{loss}} \in [0, W]$ has mean $R_Q(\tau)$, and the Section 3.1 argument rescaled by range W gives the valid p-value $p_{\tau}^{\text{exact}} = \exp(-\frac{2n_2}{W^2}(\alpha - \hat{R}_Q(\tau))^2_+)$ with $\hat{R}_Q(\tau) = \frac{1}{n} \sum_i w(X_i)\ell_i$. Theorems 1–2 then hold under Q . Two caveats matter in practice. (i) Hoeffding with range W is *loose* for importance weighting because it ignores that most weights are small; the empirical-Bernstein p-value, whose slack scales with the *weighted variance* rather than the range, is materially tighter and is what we use in experiments. (ii) The often-quoted effective sample size $n_{\text{eff}} \approx (\sum_i w_i)^2 / \sum_i w_i^2$ describes the behavior of that variance-adaptive bound, *not* of the range- W Hoeffding bound above; we report n_{eff} only alongside the empirical-Bernstein certificate, to avoid conflating the two.

B. ESTIMATED WEIGHTS

Let weights be **estimated** by any method (density-ratio estimation, a discriminative classifier, moment matching). Two problems arise: (a) the estimator $\hat{w}$ is a function of data, so treating it as fixed in a Hoeffding bound is invalid; (b) $\hat{w} \neq w$ introduces bias. We resolve (a) by **sample splitting** and (b) by an L^1 **error budget**.

Split the calibration set into independent folds $\mathcal{D}_1$ (size n_1) and $\mathcal{D}_2$ (size n_2). Estimate $\hat{w}$ on $\mathcal{D}_1$ *only*. Suppose the estimate is bounded, $0 \leq \hat{w} \leq \hat{W}$, and obeys an L^1 error budget

$$\mathbb{E}_P[|\hat{w}(X) - w(X)|] \leq \eta \quad (\text{conditionally on } \mathcal{D}_1). \quad (*)$$

Compute the weighted empirical risk on the held-out fold, $\hat{R}_Q(\tau) = \frac{1}{n_2} \sum_{i \in \mathcal{D}_2} \hat{w}(X_i)\ell_i$, and test against the **deflated** target $\alpha - \eta$:

$$p_{\tau}^Q = \exp\left(-\frac{2n_2}{\hat{W}^2}((\alpha - \eta) - \hat{R}_Q(\tau))^2_+\right). \quad (3)$$

Theorem 4 (Shift-robust validity under estimated weights). Assume $(*)$, $0 \leq \hat{w} \leq \hat{W}$, $\eta < \alpha$, and $Y \mid X$ invariant across P, Q . Then the p-value (3) is super-uniform under $H_{\tau}^Q : R_Q(\tau) > \alpha$, and the chain (or Bonferroni) certification of Section 3–Section 4 applied to $\{p_{\tau}^Q\}$ at level δ yields

$$\Pr(R_Q(\hat{\tau}) \leq \alpha) \geq 1 - \delta \quad (\text{under } Q).$$

Proof. Condition on $\mathcal{D}_1$; then $\hat{w}$ is a fixed (deterministic) function and the summands $\hat{w}(X_i)\ell_i$ for $i \in \mathcal{D}_2$ are i.i.d. in $[0, \hat{W}]$ with common mean $\mu := \mathbb{E}_P[\hat{w}\ell_{\text{loss}}]$. Decompose

$$\begin{aligned} \mu &= \mathbb{E}_P[w\ell_{\text{loss}}] + \mathbb{E}_P[(\hat{w} - w)\ell_{\text{loss}}] \\ &= R_Q(\tau) + \mathbb{E}_P[(\hat{w} - w)\ell_{\text{loss}}]. \end{aligned}$$

Since $\ell_{\text{loss}} \in [0, 1]$, the bias term is bounded by $|\mathbb{E}_P[(\hat{w} - w)\ell_{\text{loss}}]| \leq \mathbb{E}_P[|\hat{w} - w|] \leq \eta$ by $(*)$. Under H_{τ}^Q we have $R_Q(\tau) > \alpha$, hence $\mu \geq R_Q(\tau) - \eta > \alpha - \eta$. By one-sided Hoeffding with range $\hat{W}$, for any observed $\hat{R}_Q(\tau) = r \leq \alpha - \eta$,

$$\begin{aligned} \Pr(\hat{R}_Q(\tau) \leq r \mid \mathcal{D}_1) &\leq \exp\left(-\frac{2n_2}{\hat{W}^2}(\mu - r)^2\right) \\ &\leq \exp\left(-\frac{2n_2}{\hat{W}^2}((\alpha - \eta) - r)^2\right) = p_{\tau}^Q. \end{aligned}$$

So p_{τ}^Q is super-uniform under H_{τ}^Q conditionally on $\mathcal{D}_1$; since this holds for every realization of $\mathcal{D}_1$, it holds marginally. The downstream FWER arguments (Theorems 1–2) are unchanged, run conditionally on $\mathcal{D}_1$ and then marginalized, giving the stated $1 - \delta$ validity under Q . $\square$

Reading the result. The three costs are explicit. (i) Data splitting shrinks the testing fold to $n_2 < n$; we take n_1 as small as the weight estimator tolerates, since weight estimation is typically more sample-efficient than the Hoeffding slack on the testing fold. (ii) The deflation $\alpha \rightarrow \alpha - \eta$ requires $\eta < \alpha$ and makes the certificate conservative in

proportion to the weight-error budget. (iii) Range inflation $\hat{W}$ — mitigated by using the empirical-Bernstein form of (3), exactly as in Section 6.1. The budget η has two sources: a theoretical L^1 rate for the chosen density-ratio estimator under stated smoothness, or — more defensibly, since it needs no knowledge of w — a sensitivity reading: the certificate is valid for any deployment whose weight error is at most η , so the practitioner reports the largest η^* (the most severe shift) under which the target still holds.

Positioning. The two ingredients are individually standard: conditioning on a held-out fold to make a plug-in estimator a fixed function, and absorbing estimation error into a margin in the spirit of sensitivity analysis (cf. the Γ -style robustness bounds and estimated-weight treatments in the conformal-under-shift literature [8], [9]). The contribution is their combination in the LTT risk-control setting, where we have not seen the estimated-weight case stated with a finite-sample proof. We claim validity and interpretability, not tightness.

When this is useful, and when it is not. Two technical limits, both visible in Section 10.1, determine the interpretation. First, η must upper-bound the estimator's L^1 error for validity to hold; it cannot be inferred from the same calibration losses. When η is set large enough to be defensible, the deflation $\alpha - \eta$ can leave insufficient slack and make the certificate *infeasible*; in the simulation, feasibility falls toward zero once η approaches a fifth of α . The result is therefore useful when achievable risk is well below α . Second, validity is not automatic when η is under-specified: it held in one simulation only because the fitted density ratio was conservatively biased, while a deliberately anti-conservative estimator with $\mathbb{E}|\hat{w} - w| > \eta$ breaches the target in Section 10.1. Theorem 4 should therefore be read as a sensitivity statement with a measurable feasibility cost, not as a general distribution-shift solution.

VII. ALGORITHM SUMMARY

Cost: the procedure requires one forward pass collecting all exit scores per calibration point ($O(n)$ inference) and $O(nK)$ post-processing to score the chain. This calibration cost is separate from model training and from deployment latency.

VIII. RELATED WORK

- **Conformal prediction and risk control.** Split conformal prediction [10], [11], conformal risk control [12], [13], and LTT [1] provide distribution-free guarantees for data-dependent prediction sets or model configurations (see [14] for an introduction). These methods address the same statistical obstacle that invalidates naive validation thresholding: after a search over candidate thresholds, the selected configuration is no longer a fixed hypothesis. CARC uses the LTT reduction directly. The added content is the adaptive-inference instantiation: a compute-ordered candidate family, an efficiency statement for fixed-sequence certification, explicit infeasibility semantics, and a conservative joint risk–compute extension.

Algorithm 1 Compute-adaptive risk certification

Require: Multi-exit model with exits $1, \dots, L$; calibration data $\mathcal{D}$ with n points; target risk α ; confidence δ ; compute-increasing chain $\{\tau^{(1)} \preceq \dots \preceq \tau^{(K)} = \mathbf{1}\}$; optional budget B ; optional shift inputs $\{\text{weight estimator}, \eta\}$; split (δ_R, δ_C) .

- 1: **if** using the shift branch **then**
- 2: Split $\mathcal{D}$ into $\mathcal{D}_1, \mathcal{D}_2$; fit $\hat{w}$ on $\mathcal{D}_1$; set the test fold to $\mathcal{D}_2$, the target to $\alpha - \eta$, and the range to $\hat{W}$.
- 3: **else**
- 4: Set the test fold to $\mathcal{D}$, the target to α , and the range to 1.
- 5: **end if**
- 6: **for each** $\tau^{(k)}$ **do**
- 7: On the test fold compute $E_{\tau^{(k)}}(X_i)$, predictions, losses ℓ_i , costs $c_E(X_i)$, empirical risk $\hat{R}(\tau^{(k)})$ or weighted $\hat{R}_Q(\tau^{(k)})$, and empirical compute $\hat{C}(\tau^{(k)})$.
- 8: **end for**
- 9: Compute p-values p_k using Eq. (2), Hoeffding–Bentkus, empirical Bernstein, or weighted Eq. (3), at the target and range from step 0.
- 10: Certify risk with one pre-committed FWER rule: fixed-sequence on the chain, Holm step-down on a finite grid, or Bonferroni on the finite family; call the certified set $\hat{\Lambda}$.
- 11: **if** dual risk–compute control is requested **then**
- 12: Filter $\hat{\Lambda}_B = \{k \in \hat{\Lambda} : \hat{C}(\tau^{(k)}) + C_{\max} \sqrt{\ln(K/\delta_C)/(2n_{\text{test}})} \leq B\}$.
- 13: Output the certified budget-feasible configuration in $\hat{\Lambda}_B$ with smallest $\hat{C}$; if none exists, report infeasible.
- 14: **else**
- 15: Output the certified configuration in $\hat{\Lambda}$ with smallest $\hat{C}$; if none exists, report infeasible.
- 16: **end if**

Ensure: The probability of outputting a certified configuration that violates $R(\hat{\tau}) \leq \alpha$ [and $C(\hat{\tau}) \leq B$] is at most δ , under Q when the shift branch is used and its assumptions hold; an infeasible report carries no risk guarantee.

- **Multiple testing for ordered hypotheses.** Bonferroni, fixed-sequence/fallback procedures [3], Holm step-down [2], and graphical sequentially-rejective tests [4] control family-wise error for finite hypothesis families. By themselves, these procedures do not define which hypothesis order is operationally meaningful for an early-exit model. CARC fixes the order by compute before calibration, so the testing sequence matches the deployment objective: among certified policies, choose the lowest-cost one in the searched family.
- **Adaptive computation.** Adaptive-depth architectures such as BranchyNet [15], MSDNet [16], SkipNet [17], patience-based exiting [18], and LLM cascades/routers [19] reduce average inference cost by early termination or model routing; broader surveys appear in [20], [21]. These systems typically tune exit thresholds from empir-

ical validation curves. That tuning can reduce compute, but it does not certify population risk after threshold search. CARC addresses the calibration layer only: it assumes the adaptive model and candidate scores are fixed, then certifies or rejects candidate threshold policies.

- **Calibrated adaptive inference.** Closest to this setting, confident adaptive language modeling [22] calibrates early-exit thresholds for transformer decoders with an LTT-style consistency guarantee, and conformal/risk-control wrappers have been applied to selective and cascaded prediction [13], [19]. These works show that calibrated exit rules are possible for a single confidence parameter or a specific architecture. CARC differs in three deployment-relevant ways: (a) it tests policies in a compute order that coincides with the cost objective, so the certified output is the cheapest in-chain safe policy; (b) it extends from a scalar threshold to several pre-specified per-exit chains under a chain-level error split (Proposition 1); and (c) it treats an empty certified set as an explicit infeasibility report rather than an implicit fallback. The estimated-weight covariate-shift statement in Theorem 4 is included as a finite-sample sensitivity result for the same LTT risk-control problem.
- **Distribution shift.** Weighted conformal prediction [8] and sensitivity-analysis methods for conformal prediction under unknown or misspecified weights [9] address deployment distributions that differ from calibration distributions. Theorem 4 follows this conservative logic: validity requires a correct externally supplied L^1 weight-error budget, and feasibility can disappear as the budget grows. The result is a sensitivity certificate for a stated shift model, not a general robustness claim.
- **Selective prediction.** Selective risk and coverage [23] formalize systems that abstain or defer. CARC can encode selective risk through ℓ_{loss} , but its primary object is different: it certifies threshold selection over compute-adaptive policies, with compute and infeasibility reported alongside risk.

IX. IMPLEMENTATION AND EVALUATION OVERVIEW

Implemented components. The implementation follows the pipeline used in the theory. A fixed adaptive model writes cached per-exit scores, predictions, correctness indicators, and cumulative compute. The evaluator maps that cache to loss and cost matrices over a finite threshold family, computes valid p-values, applies a pre-committed FWER selector, and returns the certified set, selected threshold, feasibility status, and certificate fields for the requested (α, δ, B) . The code implements Hoeffding, Hoeffding–Bentkus, and empirical-Bernstein p-values; Bonferroni, fixed-sequence chain, Holm, and multi-chain selectors; the joint risk–compute test; the estimated-weight shift diagnostic; and cache adapters for synthetic, CIFAR branchy-network, and tabular-cascade settings.

Completed evaluations. The controlled simulation tests proof-facing properties under oracle risks. The digits tabular-cascade cache tests the same certification path on a feasible

real-cache target grid. The CIFAR-10 branchy-cache run provides a feasible neural-cache benchmark for the redesigned selector, and the CIFAR-100 branchy-cache run tests the failure mode in which the trained model cannot meet most pre-defined targets. Wall-clock latency and hardware throughput are not certified quantities in this manuscript; compute is represented by the pre-specified cost proxy in each cache.

X. EXPERIMENTS

The experiments follow the same sequence as the method: controlled validation of the certificate, expansion from scalar chains to multi-chain search, feasible real-cache evaluation, and infeasibility diagnostics. First, the controlled simulation verifies finite-sample validity and compute behavior when true risk is known. Second, the multi-chain redesign experiment tests whether richer pre-specified threshold families improve certified efficiency without losing validity. Third, real-cache experiments evaluate the same certification path on feasible tabular cascades and a feasible CIFAR-10 branchy network. Fourth, the CIFAR-100 branchy-cache experiment evaluates the opposite case, where most pre-defined targets are below the finite-pool risk frontier and the correct certified outcome is infeasibility, while also testing feasible higher-risk cells for the redesigned selector. Larger vision, language, and real-shift deployments remain outside the evidence base of this manuscript.

Figure 3 summarizes the controlled simulation, CIFAR-100 diagnostic, and synthetic shift sensitivity sweep. Table 4 reports the original positive real-cache validation, and Table 3 reports the CIFAR-10 neural-cache multi-chain comparison. The remaining multi-chain artifacts are reported in text and released as JSON result files. These artifacts support scoped claims: simulation verifies theorem-level behavior with oracle risks; multi-chain experiments test certified search over richer threshold families; digits and breast cancer validate the cache evaluator on feasible real-cache grids; CIFAR-10 evaluates the redesigned selector on a feasible neural cache; CIFAR-100 diagnoses infeasibility for a weaker branchy model and target grid; and the shift experiment is a sensitivity analysis rather than a general shift-robust deployment claim.

A. CONTROLLED-SIMULATION VERIFICATION

We use a synthetic early-exit model ($L = 5$ exits, competence increasing with depth, noisy confidences, global-threshold chain of $K = 40$ configurations) whose true per-configuration risk and cost are computable by large-sample Monte Carlo. This setting isolates the statistical guarantee: every selected policy can be checked against true risk rather than a held-out finite-pool proxy. It is not evidence about real network accuracy or hardware performance, and it does not validate Assumption M or weight estimability in deployment. The complete row-level numerical outputs are embedded in Tables 5–8.

Validity and naive failure (E1/E3). Across a grid of targets and $T = 3000$ calibration/test splits per cell, the certified chain violates $R(\hat{\tau}) \leq \alpha$ at rates of 0.0010–0.0143, all below

TABLE 2. Where CARC sits among post-calibration selection methods for adaptive inference. “DF post-selection risk”: distribution-free risk guarantee for the data-dependent selected policy. “Compute-ordered search”: the candidate family is tested in an order that coincides with the deployment cost objective. “Infeasibility reporting”: the procedure explicitly returns “no certified policy” rather than a fallback of unknown risk. Entries describe each method’s typical native form; weighted or LTT-wrapped variants can extend several rows.

Method	DF risk	Multiplicity	Ordered search	Infeasible?	Shift cert.
Naive validation tuning	No	No	Greedy (uncorrected)	No	No
Split conformal / CRC [12], [13]	Yes	Single parameter	No	Implicit	Weighted variants
LTT [1]	Yes	Yes (FWER)	Order-agnostic	Implicit	Via weighting
CARC (this paper)	Yes	Yes (FWER)	Yes (compute chain)	Yes (explicit)	Yes (sensitivity, Th. 4)

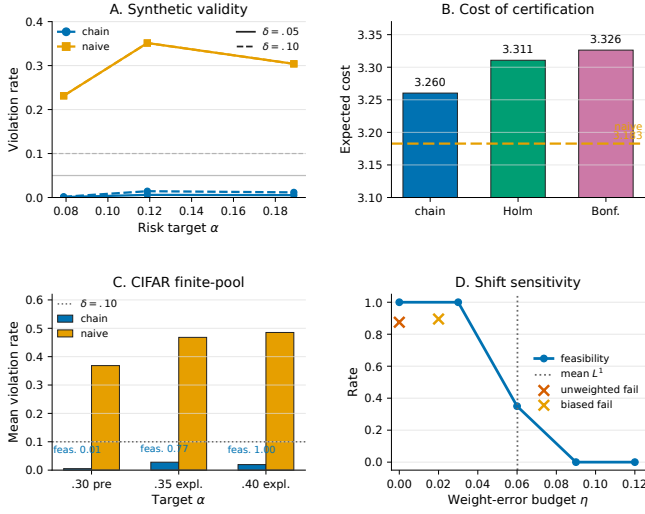


FIGURE 3. Empirical evidence summary from the completed studies. (A) In the controlled simulation, the certified chain stays below the nominal violation levels while naive validation inspection violates far above them. (B) In the representative E2 cell, the chain has lower certified expected cost than Holm or Bonferroni, and remains close to the invalid naive reference. (C) In the CIFAR-100 finite-pool diagnostic, bars show mean violation rates across the reported split cells for each target and method; vertical labels give mean feasibility, and exact cell values are in Tables 9–10. The $\alpha = 0.30$ target is pre-defined but infeasible in full-pool risk; $\alpha \in \{0.35, 0.40\}$ is exploratory. (D) Under shift, an undersized or biased weight budget can fail, while defensible η values make the estimated-weight certificate infeasible in this synthetic sweep.

$\delta \in \{0.05, 0.10\}$. The naive recipe (cheapest configuration with $\hat{R} \leq \alpha$, no multiplicity correction) violates at 0.2313–0.3513. In this oracle-risk simulation, the main failure of the uncorrected baseline is post-selection calibration error, not the underlying candidate family.

Cost of the guarantee (E2). At a representative $(\alpha, \delta) = (0.119, 0.10)$ with $T = 2000$, the certified expected cost is 3.260 for the chain, 3.311 for Holm, and 3.326 for Bonferroni; all three certified selectors satisfy the tested violation constraint. At the matched E1 target, the invalid naive policy costs 3.183. The chain therefore pays a small premium over the invalid baseline in this synthetic setting, and its advantage over Bonferroni is modest. The result supports the theoretical distinction between validity and efficiency, but it does not imply that the compute premium will be small for every trained model or target.

Joint control (E5). With $\alpha = 0.089$, $\delta = 0.10$, $n = 2000$, and $T = 1000$, a slack budget $B = 4.641$ is certified feasible in every trial, with no true risk violations and mean certified

compute UCB 3.620. A tight budget $B = 3.363$ is oracle-feasible — the cheapest truly safe configuration costs 3.313 — but the range- $C_{\max}$ compute certificate reports infeasible in every trial. This quantifies the conservativeness flagged in Section 5.

Covariate shift (E6). Under an exp-tilt shift toward harder inputs, the unweighted source-calibrated certificate violates the target at rate 0.875. The estimated-weight sweep has realized mean weight error $\|\hat{w} - w\|_1 \approx 0.060$: when $\eta \in \{0, 0.03\}$, the certificate remains feasible with 0 observed violations but lacks the assumption required by Theorem 4; at $\eta = 0.06$, feasibility drops to 0.35; and at $\eta \in \{0.09, 0.12\}$, feasibility is 0. With a deliberately downward-biased estimator and an underspecified $\eta = 0.02$ ($\|\hat{w} - w\|_1 \approx 0.351$), the certificate violates at rate $0.895 \gg \delta$. These results calibrate Theorem 4 as a sensitivity tool, not as a robustification method that can be applied without a defensible weight-error budget.

B. MULTI-CHAIN SELECTOR REDESIGN

The redesigned selector uses six pre-specified per-exit threshold chains, each with 30 levels, giving 180 candidate policies. In the synthetic E8 experiment, true risk and cost are again available from a large oracle sample. With $n = 1000$, $T = 1000$, and Hoeffding–Bentkus p-values, multi-chain CARC keeps true violation at 0–0.001 across the tested (α, δ) cells, whereas naive validation inspection violates at 0.363–0.382. Multi-chain has lower mean true cost than Holm and Bonferroni in every tested cell. Relative to scalar-chain CARC, it has higher feasibility at the strictest target and lower mean cost in five of six cells. This experiment tests Proposition 1 empirically: a richer pre-specified family can improve certified efficiency without applying a per-configuration Bonferroni correction.

The real-cache reanalysis gives the same qualitative picture, with dataset-specific limits. On the digits cache at $\alpha = 0.10$, multi-chain violation is 0.002–0.013 and mean relative cost is 2.78–2.83, compared with scalar-chain cost 6.05–6.08 and naive violation 0.442–0.542. On the breast-cancer cache, $\alpha = 0.15$ with 120 calibration examples gives multi-chain violation 0, feasibility 0.986–0.995, and mean cost 1.1–1.3, below scalar-chain and Holm/Bonferroni. The wine cache is saturated—all exits are already near-perfect—and is therefore treated as a boundary case rather than a headline result.

On CIFAR-10, the redesigned selector gives a feasible neural-cache benchmark. The branchy model reaches final-

exit test accuracy 0.8825 and a full-pool risk floor near 0.1173, so targets $\alpha \geq 0.12$ are feasible on the finite cache. With $n_{\text{cal}} = 2000$, multi-chain CARC has lower mean MAC cost than scalar-chain CARC in every tested (α, δ) cell from $\alpha = 0.12$ to 0.30; in the more stable range $\alpha = 0.15$ –0.30, it selects in 0.956–1.000 of trials, violates at 0.001–0.017, and remains cheaper than Holm and Bonferroni. Table 3 reports the $n_{\text{cal}} = 2000$ cells used for this claim.

On CIFAR-100, multi-chain does not make the old $\alpha = 0.30$ target meaningfully feasible; the trained branchy model remains too weak for that target. At feasible higher-risk targets, however, the selector improves certified efficiency relative to valid full-family baselines. At $\alpha = 0.40$ with $n_{\text{cal}} = 2000$, multi-chain is feasible in all trials, has held-out violation 0.007 at $\delta = 0.05$ and 0.011 at $\delta = 0.10$, and costs 86.59M and 85.95M MACs, respectively. The matched scalar-chain costs are 87.22M and 86.71M MACs, while Holm and Bonferroni are more expensive. These CIFAR-100 results support the redesigned selector at feasible targets but do not support low-error certified deployment for this trained CIFAR-100 model.

C. DIGITS TABULAR-CASCADE REAL-CACHE CONFIRMATION

The digits study gives a feasible real-cache check of the scalar-chain certification path. It uses the scikit-learn digits data with a three-exit tabular cascade: Gaussian naive Bayes, logistic regression, and support-vector classification, with relative exit costs 1.0, 4.0, and 12.0. The train/cache split is fixed by seed 20260615; the cache contains 899 examples, 64 features, and 10 classes. The full-pool exit accuracies are 0.863, 0.968, and 0.981, and the final-exit full-pool risk is 0.0189. Thus the pre-defined targets $\alpha \in \{0.05, 0.10, 0.15\}$ are feasible before calibration. The grid uses Hoeffding–Bentkus p-values, $T = 1000$ calibration/test partitions per cell, $\delta \in \{0.05, 0.10\}$, and calibration sizes $\{100, 300\}$. Violation rates are counted over all split attempts; infeasible reports count as no certified violation but are exposed through the feasibility column.

Across all 12 chain cells, the empirical violation rate is 0–0.051, and no cell exceeds its matching δ . The most informative contrast is at $\alpha = 0.10$: the chain selects in 0.884–1.000 of trials, violates at 0.002–0.018, and has mean cost 11.800–11.979; naive validation inspection violates at 0.486–0.571 while spending 5.14–6.09 fewer relative cost units. At $\alpha = 0.05$, the stringent target often forces the final exit. At $\alpha = 0.15$, the target is easier and the naive baseline can be safe in some cells, but Table 4 still reports the compute premium of certification. This experiment is not a claim about tabular classification performance; it tests whether the cache-based CARC pipeline reports feasible targets, infeasibility, violation rates, and compute premium without tuning on test splits. The non-neural cascade isolates the calibration layer: CARC uses the trained system only through cached scores, losses, and costs, so this study exercises the same statistical interface used by neural early-exit caches while leaving

neural-representation quality outside its scope.

D. CIFAR-100 REAL-CACHE DIAGNOSTIC

The CIFAR-100 study evaluates the infeasibility branch of CARC on a neural cache whose trained model is below the requested accuracy regime. In this setting, a valid selector should withhold certification rather than return a policy whose held-out risk remains above the target. The diagnostic uses a branchy ResNet-56-style model with four exits (after the stem and after each of three residual stages), scalar top-softmax thresholds on a $K = 100$ grid, 0–1 loss, and analytical cumulative MAC estimates. The model was trained for 80 epochs and evaluated on the CIFAR-100 test set. The per-exit test accuracies are 0.0646, 0.3040, 0.6147, and 0.6945; the corresponding cumulative exit costs are 0.44, 42.91, 84.33, and 125.75 million MACs. Because this is a finite test-pool diagnostic, the reported violation frequencies use held-out split risk as a proxy for population risk, not as an oracle truth measurement. The complete row-level numerical outputs are embedded in Tables 9–10.

Fixed CIFAR target grid. The frozen CIFAR E1 grid uses $\alpha \in \{0.05, 0.10, 0.20, 0.30\}$, $\delta \in \{0.05, 0.10\}$, calibration sizes $\{500, 2000\}$, Hoeffding–Bentkus p-values, and $T = 1000$ random calibration/test partitions per cell. The full test-pool minimum risk along the chain is 0.3055, so none of the pre-defined targets has a truly safe configuration. The certified chain therefore mostly reports infeasibility. Its violation frequency over all split attempts is 0 for $\alpha \leq 0.20$ and 0.000–0.013 at $\alpha = 0.30$, but feasibility at $\alpha = 0.30$ is only 0.000–0.013. The naive selector instead selects in 0.281–0.457 of the $\alpha = 0.30$ split attempts and violates in exactly those attempts, because the selected finite-pool policies remain above the target. This result supports the infeasibility-reporting behavior and exposes the naive failure mode; it does not support a claim that this CIFAR model enables low-error certified deployment at the pre-defined targets.

Exploratory feasible-target check. We also ran a post hoc diagnostic at $\alpha \in \{0.35, 0.40\}$ with the same δ , calibration sizes, p-values, and split count to evaluate the same cache above its finite-pool risk floor. The chain violation frequencies are 0.012–0.038, always below the corresponding δ ; naive validation tuning violates at 0.441–0.529. Chain feasibility is high for $\alpha = 0.40$ (0.994–1.000) and mixed for $\alpha = 0.35$ (0.484–0.996), matching the fact that the feasible frontier is close to the final-exit risk. Mean chain cost is 85.65–105.30 million MACs across these feasible-target cells, compared with 81.28–92.85 million MACs for the invalid naive policy. Holm and Bonferroni are more conservative: they have lower or zero violation rates but lower feasibility and higher mean cost in the same cells.

Monotonicity diagnostic. The full-pool chain risk is not perfectly nonincreasing: four adjacent threshold pairs show risk increases, with maximum increase 2×10^{-4} , while cost is nondecreasing. This small empirical violation does not affect validity, but it means the Assumption M efficiency

TABLE 3. CIFAR-10 branchy-cache multi-chain comparison at $n_{\text{cal}} = 2000$. The cache has $n = 10000$ test examples, four exits, six pre-specified chains with 30 levels each ($K = 180$ policies), Hoeffding–Bentkus p-values, $T = 1000$, and violation denominator “all”. Costs are mean analytical MACs in millions.

α	δ	MC feas.	MC viol.	MC cost	Scalar cost	Holm cost	Bonf. cost	Naive viol.
0.15	0.05	0.956	0.008	74.87	75.98	78.24	78.76	0.490
0.15	0.10	0.977	0.013	73.97	74.68	78.01	78.38	0.472
0.20	0.05	1.000	0.001	59.84	60.92	61.33	61.54	0.455
0.20	0.10	1.000	0.006	59.43	60.36	60.90	61.14	0.465
0.25	0.05	1.000	0.002	52.43	53.15	53.47	53.77	0.428
0.25	0.10	1.000	0.009	52.11	52.73	53.30	53.56	0.404
0.30	0.05	1.000	0.002	45.89	46.38	46.80	47.08	0.389
0.30	0.10	1.000	0.017	45.61	46.02	46.51	46.87	0.382

TABLE 4. Fixed-grid digits tabular-cascade real-cache confirmation. The cache uses 899 examples, $K = 101$ scalar thresholds, Hoeffding–Bentkus p-values, $T = 1000$, violation denominator “all”, and relative costs [1, 4, 12]. The certified compute premium is mean chain cost minus mean naive cost in the same relative cost units.

α	δ	n_{cal}	Chain feas.	Chain viol.	95% CI	Chain cost	Naive viol. / premium
0.05	0.05	100	0.139	0.000	[0.000,0.004]	12.000	0.045 / 0.483
0.05	0.05	300	0.832	0.000	[0.000,0.004]	12.000	0.000 / 0.000
0.05	0.10	100	0.413	0.000	[0.000,0.004]	12.000	0.047 / 0.503
0.05	0.10	300	0.924	0.000	[0.000,0.004]	12.000	0.000 / 0.000
0.10	0.05	100	0.884	0.007	[0.003,0.014]	11.915	0.571 / 6.092
0.10	0.05	300	1.000	0.002	[0.000,0.007]	11.979	0.497 / 5.281
0.10	0.10	100	0.964	0.018	[0.011,0.028]	11.800	0.522 / 5.444
0.10	0.10	300	1.000	0.004	[0.001,0.010]	11.958	0.486 / 5.142
0.15	0.05	100	1.000	0.000	[0.000,0.004]	10.115	0.000 / 8.701
0.15	0.05	300	1.000	0.034	[0.024,0.047]	4.985	0.049 / 3.971
0.15	0.10	100	1.000	0.001	[0.000,0.006]	8.803	0.001 / 7.330
0.15	0.10	300	1.000	0.051	[0.038,0.067]	4.049	0.051 / 3.024

interpretation should be read as a diagnostic rather than a confirmed model property on this cache.

XI. LIMITATIONS

- **Calibration assumptions.** The base validity result assumes i.i.d. calibration examples drawn i.i.d. from the deployment distribution. Under covariate shift, Theorem 4 additionally requires an externally justified L^1 weight-error budget η ; if that budget is too small or the shift is not represented by the weighting model, the certificate can fail.
- **Loss assumptions.** The concentration bounds require bounded losses. Unbounded or heavy-tailed risk functionals require different concentration arguments, such as truncation or variance-adaptive bounds under additional moment conditions.
- **Search scope.** Cost-optimality is only within the chosen finite candidate family. A scalar threshold chain is a one-dimensional subset of the full per-exit threshold space, and neither the chain nor the implemented Holm baseline exhausts the continuum of possible policies.
- **Multi-chain design.** Multi-chain CARC broadens the searched family only over chains fixed before calibration. Validity does not require those chains to be efficient, but efficiency does. Poorly chosen chains can miss cheaper safe policies, and choosing chains after inspecting calibration risks would introduce multiplicity outside Proposition 1.
- **Compute model.** The certified compute functional uses the pre-specified cost proxy c_ℓ . In the CIFAR diagnostics this proxy is analytical MAC count, not measured wall-

clock latency. Hardware latency under batching, memory effects, and accelerator scheduling must be evaluated separately before deployment.

- **Efficiency assumptions.** Assumption M affects the efficiency interpretation, not validity. If risk is not monotone along the compute chain, the selector remains protected by FWER control, but the chain-order cost bound should not be read as an empirical efficiency guarantee. The reported monotonicity diagnostic is necessary but not sufficient for the true-risk condition.
- **Empirical scope.** The real-cache evidence consists of feasible tabular cascades, a saturated wine boundary case, a feasible CIFAR-10 branchy CNN, and a CIFAR-100 diagnostic on a weaker branchy CNN. These caches test calibration behavior rather than benchmark superiority. The CIFAR-100 fixed target grid is mostly infeasible because the final-exit test risk is 0.3055, and the feasible CIFAR-100 targets are exploratory.
- **Shift certificate.** Theorem 4 is a conservative sufficient condition. When η is large enough to be defensible, the deflated target can make certification infeasible; when η is too small, the target can be breached. The budget must therefore come from an external estimator guarantee or be reported as a sensitivity parameter.
- **Single-use protocol.** The guarantee is for one pre-specified run of the procedure. Sweeping multiple $(\alpha, \delta, \text{chain})$ choices on the same calibration set and then selecting the most favorable outcome introduces multiplicity outside the stated guarantee.

- **Future validation.** The next empirical step is evaluation under a standardized protocol on larger early-exit models, language-model cascades, measured hardware latency, and non-synthetic distribution shifts. Each study should report infeasibility, compute premium, and violation rates, rather than only the cheapest selected policy.

XII. ENGINEERING AND DEPLOYMENT CONSIDERATIONS

The intended deployment role of CARC is narrow: it is a calibration layer for a fixed adaptive inference system. It can certify a threshold policy, report that no candidate certifies, or quantify the compute premium of certification relative to an uncorrected baseline. It does not improve the trained model, repair poor final-exit accuracy, or guarantee latency under hardware effects not represented in c_ℓ . The main operational risk is over-trust: an infeasible report, a finite-pool diagnostic, or an exploratory feasible-target check can be mistaken for a population-level deployment guarantee. The manuscript mitigates this risk by stating assumptions at the point of use, separating feasibility from validity, and requiring (α, δ) , the chain, and the split protocol to be fixed before calibration risks are inspected.

A worked deployment read. Table 4 can be read as an operator would use it. Suppose the target is at most 10% error with 95% confidence ($\alpha = 0.10$, $\delta = 0.05$) and 300 calibration points are available. CARC certifies a policy in every trial at mean relative cost 11.98 while violating the target in only 0.2% of calibration draws; the uncorrected baseline is about 5.3 cost units cheaper but violates the 10% target in roughly half of the draws. That gap is the measured compute premium for certification in this cell. Tightening the target to $\alpha = 0.05$ with only 100 calibration points instead gives feasibility 0.139: CARC reports infeasibility in most splits, indicating that this system, at this calibration size, cannot certify 5% error. The operator must then collect more calibration data, relax the target, improve the model, or decline deployment of the uncertified policy. Relaxing the target to $\alpha = 0.15$ moves the certified policy to a much cheaper exit (mean cost 4.05), so the certified operating point tracks the requested risk level.

XIII. CONCLUSION

This paper formulates threshold selection for compute-adaptive inference as finite-family risk control. By instantiating LTT on compute-ordered candidate chains, CARC separates three quantities that validation tuning often conflates: finite-sample risk validity, feasibility of the requested target for the trained system, and compute efficiency of the certified policy. The theory proves distribution-free validity for certified selectors, characterizes the finite-sample compute band of the fixed-sequence chain under monotone risk, extends the selector to several pre-specified chains with chain-level error allocation, adds a conservative joint risk–compute certificate, and gives an estimated-weight sufficient condition for covariate shift.

The empirical evidence supports three scoped conclusions. First, in controlled simulations with oracle risks, certified selection respects the tested violation levels while naive validation thresholding violates substantially. Second, expanding from a scalar chain to pre-specified multiple chains can reduce certified cost relative to valid full-family baselines while preserving the same error-control principle. Third, infeasibility is an informative output: the digits and CIFAR-10 caches contain feasible targets, whereas the CIFAR-100 fixed grid mostly does not because the trained final exit has test risk 0.3055. These results support CARC as a calibration layer for fixed adaptive systems, not as a training method or a general guarantee that any early-exit network can meet a requested target.

The main limitations are structural. Validity assumes i.i.d. calibration data unless the covariate-shift certificate is invoked with a defensible η ; the compute-efficiency interpretation of Theorem 2 depends on the chosen chain and Assumption M; the joint compute certificate is conservative because it uses range-based concentration; and the real-cache evidence is limited to tabular cascades plus CIFAR-10/CIFAR-100 branchy CNN caches. Future work should evaluate larger early-exit networks, language-model cascades, measured hardware latency, and non-synthetic distribution shifts under the same standardized evaluation protocol, and should develop tighter variance-adaptive or hardware-aware compute certificates without weakening finite-sample risk control.

XIV. DATA AND CODE AVAILABILITY

The current reference implementation covers the p-values, certification procedures, scalar-chain and multi-chain selectors, threshold-family builders, controlled-simulation scripts, CIFAR cache adapter, tabular-cascade adapter, and cache-based real-data evaluator. The digits scalar-chain result is summarized in Table 4; the CIFAR-10 multi-chain result is summarized in Table 3; the synthetic and CIFAR-100 scalar-chain row-level results are embedded in Appendix B. The corresponding JSON artifacts are stored under `results/`: `digits_tabular_real_cache_eval.json`, `e8_multichain_selector.json`, `digits_tabular_multichain_eval.json`, `breast_cancer_tabular_multichain_eval.json`, `wine_tabular_multichain_eval.json`, `cifar10_branchy_multichain_eval.json`, `cifar_branchy_real_cache_eval.json`, `cifar_branchy_real_cache_eval_exploratory.json`, and `cifar_branchy_multichain_eval.json`. All (α, δ, B) grids, seeds, split counts, and pass criteria used for reported experiments are specified in the source files or in `preregistration.md`; exploratory CIFAR-100 cells are labeled separately. The reference implementation, cache adapters, and every JSON artifact listed above are provided in the supplementary material accompanying this submission, so all reported headline numbers are independently reproducible from the

released code and caches. The public repository URL and the archival DOI assigned at acceptance will be inserted in the camera-ready version.

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REFERENCES

- [1] A. N. Angelopoulos, S. Bates, E. J. Candès, M. I. Jordan, and L. Lei, "Learn then test: Calibrating predictive algorithms to achieve risk control," *The Annals of Applied Statistics*, vol. 19, no. 2, Jun. 2025. [Online]. Available: <https://doi.org/10.1214/24-AOAS1998>
- [2] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979. [Online]. Available: <https://www.jstor.org/stable/4615733>
- [3] B. L. Wiens, "A fixed sequence bonferroni procedure for testing multiple endpoints," *Pharmaceutical Statistics*, vol. 2, no. 3, pp. 211–215, Jul. 2003. [Online]. Available: <https://doi.org/10.1002/pst.64>
- [4] F. Bretz, W. Maurer, W. Brannath, and M. Posch, "A graphical approach to sequentially rejective multiple test procedures," *Statistics in Medicine*, vol. 28, no. 4, pp. 586–604, Dec. 2008. [Online]. Available: <https://doi.org/10.1002/sim.3495>
- [5] W. Hoeffding, "Probability inequalities for sums of bounded random variables," *Journal of the American Statistical Association*, vol. 58, no. 301, pp. 13–30, Mar. 1963. [Online]. Available: <https://doi.org/10.1080/01621459.1963.10500830>
- [6] V. Bentkus, "On hoeffding's inequalities," *The Annals of Probability*, vol. 32, no. 2, Apr. 2004. [Online]. Available: <https://doi.org/10.1214/009117904000000360>
- [7] A. Maurer and M. Pontil, "Empirical bernstein bounds and sample variance penalization," in *Proceedings of the Conference on Learning Theory (COLT)*, 2009.
- [8] R. J. Tibshirani, R. F. Barber, E. J. Candès, and A. Ramdas, "Conformal prediction under covariate shift," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
- [9] Y. Jin, Z. Ren, and E. J. Candès, "Sensitivity analysis of individual treatment effects: A robust conformal inference approach," *Proceedings of the National Academy of Sciences*, vol. 120, no. 6, Feb. 2023. [Online]. Available: <https://doi.org/10.1073/pnas.2214889120>
- [10] V. Vovk, A. Gammernan, and G. Shafer, *Algorithmic Learning in a Random World*. Springer International Publishing, 2022. [Online]. Available: <https://doi.org/10.1007/978-3-031-06649-8>
- [11] J. Lei, M. G'Sell, A. Rinaldo, R. J. Tibshirani, and L. Wasserman, "Distribution-free predictive inference for regression," *Journal of the American Statistical Association*, vol. 113, no. 523, pp. 1094–1111, Jun. 2018. [Online]. Available: <https://doi.org/10.1080/01621459.2017.1307116>
- [12] S. Bates, A. Angelopoulos, L. Lei, J. Malik, and M. Jordan, "Distribution-free, risk-controlling prediction sets," *Journal of the ACM*, vol. 68, no. 6, pp. 1–34, Sep. 2021. [Online]. Available: <https://doi.org/10.1145/3478535>
- [13] A. N. Angelopoulos, S. Bates, A. Fisch, L. Lei, and T. Schuster, "Conformal risk control," in *The Twelfth International Conference on Learning Representations*, 2024. [Online]. Available: <https://openreview.net/forum?id=33XGfHLtZg>
- [14] A. N. Angelopoulos and S. Bates, "Conformal prediction: A gentle introduction," *Foundations and Trends in Machine Learning*, vol. 16, no. 4, pp. 494–591, 2023. [Online]. Available: <https://doi.org/10.1561/9781638281597>
- [15] S. Teerapittayanon, B. McDanel, and H. T. Kung, "Branchynet: Fast inference via early exiting from deep neural networks," in *2016 23rd International Conference on Pattern Recognition (ICPR)*. IEEE, Dec. 2016, pp. 2464–2469. [Online]. Available: <https://doi.org/10.1109/ICPR.2016.7900006>
- [16] G. Huang, D. Chen, T. Li, F. Wu, L. van der Maaten, and K. Q. Weinberger, "Multi-scale dense networks for resource efficient image classification," in *International Conference on Learning Representations (ICLR)*, 2018.
- [17] X. Wang, F. Yu, Z.-Y. Dou, T. Darrell, and J. E. Gonzalez, "Skipnet: Learning dynamic routing in convolutional networks," in *Computer Vision – ECCV 2018*. Springer International Publishing, 2018, pp. 420–436. [Online]. Available: https://doi.org/10.1007/978-3-030-01261-8_25
- [18] W. Zhou, C. Xu, T. Ge, J. McAuley, K. Xu, and F. Wei, "Bert loses patience: Fast and robust inference with early exit," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- [19] L. Chen, M. Zaharia, and J. Zou, "Frugalgpt: How to use large language models while reducing cost and improving performance," 2023. [Online]. Available: <https://arxiv.org/abs/2305.05176>
- [20] Y. Han, G. Huang, S. Song, L. Yang, H. Wang, and Y. Wang, "Dynamic neural networks: A survey," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 11, pp. 7436–7456, 2022.
- [21] S. Scardapane, M. Scarpiniti, E. Baccarelli, and A. Uncini, "Why should we add early exits to neural networks?" *Cognitive Computation*, vol. 12, no. 5, pp. 954–966, 2020. [Online]. Available: <https://doi.org/10.1007/s12559-020-09734-4>
- [22] T. Schuster, A. Fisch, J. Gupta, M. Dehghani, D. Bahri, V. Q. Tran, Y. Tay, and D. Metzler, "Confident adaptive language modeling," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- [23] Y. Geifman and R. El-Yaniv, "Selective classification for deep neural networks," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

APPENDIX A PROOFS

Throughout, losses are in $[0, 1]$, calibration data are n i.i.d. draws from P , and $\epsilon_n(\beta) := \sqrt{\ln(1/\beta)/(2n)}$. A p-value p for a null H is *valid* (super-uniform) if $\Pr_H(p \leq u) \leq u$ for all $u \in (0, 1)$.

A. VALIDITY OF THE Hoeffding P-VALUE (EQ. 2)

Claim. $p_\tau = \exp(-2n(\alpha - \hat{R}(\tau))_+^2)$ is valid for the null $H_\tau : R(\tau) > \alpha$ of Section 3; super-uniformity is established over its closure $R(\tau) \geq \alpha$, the supremum of the rejection probability being attained at the boundary $R(\tau) = \alpha$.

Proof. Fix τ and write $R = R(\tau)$, $\hat{R} = \hat{R}(\tau)$. For $u \in (0, 1)$, $p_\tau \leq u \iff (\alpha - \hat{R})_+ \geq \epsilon_n(u) \iff \hat{R} \leq \alpha - \epsilon_n(u)$ (the event is empty if $\alpha - \epsilon_n(u) < 0$, making the probability $0 \leq u$ trivially). Under H_τ , $R \geq \alpha$, so

$$\Pr_{H_\tau}(p_\tau \leq u) = \Pr(\hat{R} \leq \alpha - \epsilon_n(u)) \leq \Pr(\hat{R} \leq R - \epsilon_n(u)) \leq e^{-2n\epsilon_n(u)^2} = u,$$

where the second inequality uses $\alpha \leq R$ and the third is the lower-tail Hoeffding inequality for the mean of n i.i.d. $[0, 1]$ variables. Hence p_τ is super-uniform. The supremum over the composite null is attained at the boundary $R = \alpha$, as used above. $\square$

B. BONFERRONI FWER (THEOREM 1)

Claim. With $\hat{\Lambda}_{\text{Bonf}} = \{\tau : p_\tau \leq \delta/|\Lambda|\}$ and valid $\{p_\tau\}$, $\Pr(\exists \tau \in \hat{\Lambda}_{\text{Bonf}} : R(\tau) > \alpha) \leq \delta$.

Proof. Let $\mathcal{N} = \{\tau : R(\tau) > \alpha\}$ be the true nulls. The bad event is $\bigcup_{\tau \in \mathcal{N}} \{p_\tau \leq \delta/|\Lambda|\}$. By validity, $\Pr(p_\tau \leq \delta/|\Lambda|) \leq \delta/|\Lambda|$ for each $\tau \in \mathcal{N}$, so by the union bound the probability is at most $|\mathcal{N}| \cdot \delta/|\Lambda| \leq |\Lambda| \cdot \delta/|\Lambda| = \delta$. Restricting any selection rule to the complement of the bad event yields a safe certified output. If the certified set is empty, the procedure reports infeasibility and makes no certified-risk claim about any operational fallback. $\square$

C. FIXED-SEQUENCE FWER (THEOREM 2(A))

Claim. Testing a pre-specified order $\tau^{(K)}, \dots, \tau^{(1)}$, certifying while $p_{\tau^{(k)}} \leq \delta$ and stopping at the first failure, controls the probability of certifying any unsafe configuration at δ , independent of K .

Proof. Let j^* be the position of the first true null in the testing order (the first tested $\tau^{(k)}$ with $R(\tau^{(k)}) > \alpha$); if there is none, no unsafe config exists and the error probability is 0. The procedure certifies a configuration only if it certifies everything earlier in the order, so it certifies an unsafe configuration only if it certifies the one at position j^* — which requires p at that position $\leq \delta$. Because the order is fixed *before* seeing the data, that hypothesis is a fixed true null, and by validity $\Pr(p \leq \delta) \leq \delta$. Thus the probability of certifying any unsafe configuration is at most δ , with no dependence on K (no correction is applied). $\square$

D. TWO-SIDED COST BOUND (THEOREM 2(B))

Stated and proved in full in the dedicated subsection below (kept separate for length). It establishes $\Pr(k_0 \leq k^* \leq k_1) \geq 1 - 2\delta$ with band $\Delta_n = \epsilon_n(\delta) + \epsilon_n(\delta/K)$, so validity carries no K -penalty while the search cost is an $O(\sqrt{\ln(K/\delta)/n})$ efficiency band.

E. JOINT RISK-COMPUTE CONTROL (THEOREM 3)

Claim. With $\delta = \delta_R + \delta_C$, risk certified at δ_R (Section A.2–A.4) and the uniform compute bound $U_C(\tau; \delta_C/K) = \hat{C}(\tau) + C_{\max} \sqrt{\ln(K/\delta_C)/(2n)}$ over the K chain points, the probability that the dual selector outputs a configuration violating either $R(\hat{\tau}) \leq \alpha$ or $C(\hat{\tau}) \leq B$ is at most δ . If the budget-certified set is empty, the selector reports infeasibility.

Proof. Define $\mathcal{A} = \{\tau \in \hat{\Lambda} : R(\tau) \leq \alpha\}$ (the risk-certification event) and $\mathcal{B} = \{\tau^k : C(\tau^k) \leq U_C(\tau^k; \delta_C/K)\}$. For $\mathcal{B}$: $c_{E_\tau(X)} \in [0, C_{\max}]$, so for each fixed k , upper-tail Hoeffding gives $\Pr(C(\tau^k) > U_C(\tau^k; \delta_C/K)) \leq \delta_C/K$; a union bound over the K (data-independent) points gives $\Pr(\mathcal{B}^c) \leq \delta_C$. By construction $\Pr(\mathcal{A}^c) \leq \delta_R$. On $\mathcal{A} \cap \mathcal{B}$, the selected $\hat{\tau} \in \hat{\Lambda}$ has $R(\hat{\tau}) \leq \alpha$, and $\hat{\tau} \in \hat{\Lambda}_B$ forces $C(\hat{\tau}) \leq U_C(\hat{\tau}; \delta_C/K) \leq B$. Boole: $\Pr((\mathcal{A} \cap \mathcal{B})^c) \leq \delta_R + \delta_C = \delta$. The denominator K is deterministic, so no data-dependent correction is involved. $\square$

F. ESTIMATED-WEIGHT SHIFT CERTIFICATE (THEOREM 4)

Claim. Under $(\star)$ $\mathbb{E}_P[\hat{w} - w] \leq \eta$ (conditionally on $\mathcal{D}_1$), $0 \leq \hat{w} \leq \hat{W}$, $\eta < \alpha$, and $Y \mid X$ invariant, the p-value (3) is valid for $H_\tau^Q : R_Q(\tau) > \alpha$, and certification at level δ gives $\Pr(R_Q(\hat{\tau}) \leq \alpha) \geq 1 - \delta$ under Q .

Proof. Condition on $\mathcal{D}_1$. Then $\hat{w}$ is a fixed measurable function, and for $i \in \mathcal{D}_2$ the variables $\hat{w}(X_i)\ell_i$ are i.i.d. on $[0, \hat{W}]$ with common mean $\mu := \mathbb{E}_P[\hat{w}\ell_{\text{loss}}]$. Decompose $\mu = R_Q(\tau) + \mathbb{E}_P[(\hat{w} - w)\ell_{\text{loss}}]$, using $R_Q(\tau) = \mathbb{E}_P[w\ell_{\text{loss}}]$ (which holds because $Y \mid X$ is invariant). Since $\ell_{\text{loss}} \in [0, 1]$, the bias term is bounded: $|\mathbb{E}_P[(\hat{w} - w)\ell_{\text{loss}}]| \leq \mathbb{E}_P[\hat{w} - w] \leq \eta$. Under H_τ^Q , $R_Q(\tau) > \alpha$, hence $\mu > \alpha - \eta$. For observed $\hat{R}_Q = r \leq \alpha - \eta$, the range- $\hat{W}$ Hoeffding inequality gives

$$\Pr(\hat{R}_Q \leq r \mid \mathcal{D}_1) \leq \exp\left(-\frac{2n_2}{\hat{W}^2}(\mu - r)^2\right) \leq \exp\left(-\frac{2n_2}{\hat{W}^2}((\alpha - \eta) - r)^2\right) = p_\tau^Q.$$

So p_τ^Q is super-uniform under H_τ^Q conditionally on $\mathcal{D}_1$; since this holds for every realization of $\mathcal{D}_1$, it holds marginally (integrate over $\mathcal{D}_1$). The FWER arguments of A.2–A.4 then run conditionally on $\mathcal{D}_1$ and marginalize, giving $1 - \delta$ validity under Q . The empirical-Bernstein variant replaces the range- $\hat{W}$ Hoeffding step with A.7 applied on $[0, \hat{W}]$, unchanged otherwise. $\square$

G. Hoeffding–Bentkus AND EMPIRICAL-BERNSTEIN P-VALUES

Hoeffding–Bentkus. For $H : R \geq \alpha$ with $\hat{R}$ the mean of n i.i.d. $[0, 1]$ losses, the p-value $p^{\text{HB}} = \min\{\exp(-n \text{kl}(\hat{R}, \alpha)), e \Pr(\text{Bin}(n, \alpha) \leq \lceil n\hat{R} \rceil)\}$ (with kl the binary KL divergence, and $p^{\text{HB}} = 1$ if $\hat{R} \geq \alpha$) is valid; the first term is the Cramér–Chernoff (relative-entropy) bound and the second the Bentkus inequality, each super-uniform under H , so their minimum is super-uniform. We use this as the default in experiments and rely on the concentration inequalities used in [12], [1], and [13], together with Bentkus’s bound [6].

Empirical Bernstein (as implemented). Let $\hat{V}_n$ be the unbiased sample variance of the n losses (range ρ ; $\rho = 1$ unweighted, $\rho = \hat{W}$ weighted). The one-sided empirical-Bernstein inequality in [7] states that, for every $\beta \in (0, 1)$,

$$\Pr\left(R - \hat{R} \geq \sqrt{\frac{2\hat{V}_n \ln(1/\beta)}{n}} + \frac{7\rho \ln(1/\beta)}{3(n-1)}\right) \leq \beta. \quad (\text{MP})$$

Define, for observed $(\hat{R}, \hat{V}_n)$ with $\hat{R} < \alpha$, the value $u^* \geq 0$ solving

$$\alpha - \hat{R} = \sqrt{\frac{2\hat{V}_n}{n}} \sqrt{u^*} + \frac{7\rho}{3(n-1)} u^*$$

(the right side is continuous and strictly increasing from 0 to ∞ in u^* , so the root is unique; in closed form, with $a = \sqrt{2\hat{V}_n/n}$, $b = 7\rho/(3(n-1))$, $c = \alpha - \hat{R}$, $\sqrt{u^*} = (-a + \sqrt{a^2 + 4bc})/(2b)$). Set $p^{\text{EB}} = \exp(-u^*)$ (and $p^{\text{EB}} = 1$ if $\hat{R} \geq \alpha$).

Claim. p^{EB} is valid for $H : R \geq \alpha$.

Proof. Because the deviation $D(\beta) := \sqrt{2\hat{V}_n \ln(1/\beta)/n} + 7\rho \ln(1/\beta)/(3(n-1))$ is strictly increasing in $\ln(1/\beta)$, the definition of u^* gives the equivalence $p^{\text{EB}} \leq \beta \iff u^* \geq \ln(1/\beta) \iff \alpha - \hat{R} \geq D(\beta)$. Under H , $R \geq \alpha$, so $\{\alpha - \hat{R} \geq D(\beta)\} \subseteq \{R - \hat{R} \geq D(\beta)\}$, and by (MP),

$$\Pr_H(p^{\text{EB}} \leq \beta) = \Pr_H(\alpha - \hat{R} \geq D(\beta)) \leq \Pr_H(R - \hat{R} \geq D(\beta)) \leq \beta.$$

Hence p^{EB} is super-uniform. Validity holds with $\hat{V}_n$ random because (MP) is itself an empirical-variance inequality, valid simultaneously over the realized $\hat{V}_n$. $\square$

(The reference implementation computes exactly this u^* ; the unit tests confirm $\Pr(p \leq \beta) \leq \beta$ under $R = \alpha$ by Monte Carlo for all three p-values.)

H. FULL PROOF OF THE TWO-SIDED COST BOUND (THEOREM 2(B))

Standing notation.

(N1) Calibration data $\mathcal{D} = \{(X_i, Y_i)\}_{i=1}^n$ i.i.d. from P .

(N2) Chain $\tau^{(1)}, \dots, \tau^{(K)}$ with deterministically nondecreasing compute $C_1 \leq \dots \leq C_K$, where $C_k := C(\tau^{(k)})$ (Section 4.1: raising the global threshold can only push each input to a deeper exit, so the per-input cost $c_{E_{\tau^{(k)}}(\cdot)}$ is pointwise nondecreasing in k , hence so is its expectation).

(N3) Per-point losses $\ell_{ik} := \ell_{\text{loss}}(\hat{Y}_{\tau^{(k)}}(X_i), Y_i) \in [0, 1]$; $\hat{R}_k := \frac{1}{n} \sum_i \ell_{ik}$, $R_k := \mathbb{E}[\hat{R}_k] = R(\tau^{(k)})$.

(N4) Slack $\epsilon_n(\beta) := \sqrt{\ln(1/\beta)/(2n)}$. With the Hoeffding p-value $p_k = \exp(-2n(\alpha - \hat{R}_k)_+^2)$, the certification test $p_k \leq \delta$ is exactly $\hat{R}_k \leq \alpha - \epsilon_n(\delta)$.

The fixed-sequence walk from expensive to cheap certifies a suffix of the compute order; equivalently

$$k^* = \min \left\{ k : \hat{R}_j \leq \alpha - \epsilon_n(\delta) \text{ for all } j \geq k \right\},$$

with the convention $k^* := K + 1$ (report infeasible) if the set is empty. Output $\hat{\tau} = \tau^{(k^*)}$ only when $k^* \leq K$. For cost-accounting diagnostics one may record $\tau^{(K)}$ as an operational fallback in infeasible cells, but that fallback is not a certified output unless it separately certifies. Let $k_0 := \min\{k : R_k \leq \alpha\}$ be the cheapest truly-safe index ($k_0 := K + 1$ if no config is safe).

Lemma A.4.1 (Hoeffding, both tails). For each fixed k and any $t \geq 0$,

$$\Pr(\hat{R}_k \leq R_k - t) \leq e^{-2nt^2}, \quad \Pr(\hat{R}_k \geq R_k + t) \leq e^{-2nt^2}.$$

Proof. The $\{\ell_{ik}\}_{i=1}^n$ are i.i.d. in $[0, 1]$ with mean R_k ; apply Hoeffding’s inequality to each tail. $\square$

No independence across configurations k is used anywhere below: every aggregation over k is a union (Boole) bound, which holds for arbitrarily dependent events.

Input — validity (Appendix A.3, restated). Because the testing order is fixed a priori and the walk stops at the first non-rejection, the event $\mathcal{U} := \{k^* \leq K \text{ and } R_{k^*} > \alpha\}$ — "certify and output an unsafe config" — satisfies $\Pr(\mathcal{U}) \leq \delta$. Write $\mathcal{V} := \mathcal{U}^c$, so $\Pr(\mathcal{V}) \geq 1 - \delta$, and on $\mathcal{V}$ the method either reports infeasible or outputs a config with $R_{k^*} \leq \alpha$.

Proposition A.4.2 (safety floor — lower bound).

$$\Pr\left(C(\hat{\tau}) \geq \min_{k: R_k \leq \alpha} C_k\right) \geq 1 - \delta.$$

Under Assumption M ($R_1 \geq \dots \geq R_K$) this is equivalently $\Pr(k^* \geq k_0) \geq 1 - \delta$.

Proof. Work on $\mathcal{V}$. If the method certifies ($k^* \leq K$), then $R_{k^*} \leq \alpha$, so $\tau^{(k^*)}$ is safe and $C(\hat{\tau}) = C_{k^*} \geq \min_{k: R_k \leq \alpha} C_k$ by definition of the minimum. If it reports infeasible, there is no certified output; for diagnostic cost accounting, assigning the infeasible cell to $\tau^{(K)}$ makes the displayed lower-bound inequality trivial because C_K is the largest compute, but it does not create a risk guarantee. Under Assumption M the safe set $\{k : R_k \leq \alpha\} = \{k_0, \dots, K\}$ is a suffix and C_k is nondecreasing, so $\min_{k \geq k_0} C_k = C_{k_0}$; moreover any $k < k_0$ has $R_k \geq R_{k_0-1} > \alpha$, so a safe output forces $k^* \geq k_0$. $\square$

Proposition A.4.3 (efficiency ceiling — upper bound). For any $\beta \in (0, 1)$ put

$$\Delta_n(\beta) := \epsilon_n(\delta) + \epsilon_n(\beta/K), \quad k_1(\beta) := \min\{k : R_j \leq \alpha - \Delta_n(\beta) \text{ for all } j \geq k\}.$$

Then $\Pr(k^* \leq k_1(\beta)) \geq 1 - \beta$.

Proof. If every $j \geq k_1(\beta)$ certifies — i.e. $\hat{R}_j \leq \alpha - \epsilon_n(\delta)$ — then by definition $k^* \leq k_1(\beta)$. Hence

$$\Pr(k^* > k_1(\beta)) \leq \Pr(\exists j \geq k_1(\beta) : \hat{R}_j > \alpha - \epsilon_n(\delta)).$$

Fix $j \geq k_1(\beta)$. By the defining property of $k_1(\beta)$, $R_j \leq \alpha - \Delta_n(\beta) = \alpha - \epsilon_n(\delta) - \epsilon_n(\beta/K)$, so $\alpha - \epsilon_n(\delta) \geq R_j + \epsilon_n(\beta/K)$, and the upper tail of Lemma A.4.1 with $t = \epsilon_n(\beta/K)$ gives

$$\Pr(\hat{R}_j > \alpha - \epsilon_n(\delta)) \leq \Pr(\hat{R}_j > R_j + \epsilon_n(\beta/K)) \leq e^{-2n \epsilon_n(\beta/K)^2} = \beta/K.$$

A union bound over the at most K indices $j \in \{k_1(\beta), \dots, K\}$ gives $\Pr(k^* > k_1(\beta)) \leq K \cdot (\beta/K) = \beta$. $\square$

(Proposition A.4.3 uses neither Assumption M nor cross-config independence.)

Theorem A.4.4 (two-sided cost bound). With $\beta = \delta$, write

$$\Delta_n := \epsilon_n(\delta) + \epsilon_n(\delta/K) = O\left(\sqrt{\frac{\ln(K/\delta)}{n}}\right), \quad k_1 := k_1(\delta).$$

Then on an event of probability at least $1 - 2\delta$,

$$k_0 \leq k^* \leq k_1, \quad \text{and} \quad 0 \leq C(\hat{\tau}) - C_{k_0} \leq C_{k_1} - C_{k_0}.$$

Under Assumption M the thresholds collapse to $k_0 = \min\{k : R_k \leq \alpha\}$ and $k_1 = \min\{k : R_k \leq \alpha - \Delta_n\}$ (the "for all $j \geq k$ " conditions reduce to single-index conditions because R_k is nonincreasing), so the excess compute $C_{k_1} - C_{k_0}$ is exactly the compute spanned by configurations whose true risk lies in the boundary band $(\alpha - \Delta_n, \alpha]$.

Proof. Intersect the events of Propositions A.4.2 and A.4.3 (with $\beta = \delta$); by Boole's inequality the intersection has probability $\geq 1 - 2\delta$. There $k_0 \leq k^* \leq k_1$, and the compute inequalities follow because C_k is nondecreasing in k . The Assumption-M simplification is immediate from monotone R_k . $\square$

Corollary A.4.5 (consistency and rate). Suppose along the band the chain relates compute to risk Lipschitz-ly: $\exists \Lambda_C < \infty$ with $C_{k_1} - C_{k_0} \leq \Lambda_C (R_{k_0} - R_{k_1}) \leq \Lambda_C \Delta_n$. Then with probability $\geq 1 - 2\delta$,

$$C(\hat{\tau}) - \min_{k: R_k \leq \alpha} C_k \leq \Lambda_C \Delta_n = O\left(\sqrt{\frac{\ln(K/\delta)}{n}}\right) \xrightarrow{n \rightarrow \infty} 0.$$

Proof. Immediate from Theorem A.4.4 and the assumed Lipschitz relation. $\square$

Remarks.

(i) *Where each ingredient enters.* Validity (Prop. A.4.2) is K -free: it is the level- δ fixed-sequence guarantee. The $\ln K$ enters only the efficiency band Δ_n (Prop. A.4.3), through the union bound over the certified suffix. This is the precise sense of "the cost of a length- K search is paid in efficiency, not validity."

(ii) *Decoupled confidences.* Validity holds at level δ and the ceiling at an independent level β ; reporting the sandwich at $1 - \delta - \beta$ lets one buy a tighter efficiency statement (smaller β , wider band) without touching validity.

(iii) *Role of Assumption M.* Used only to turn the safety floor into the index form $k^* \geq k_0$ and to collapse k_0, k_1 into clean risk thresholds. The probabilistic content of Props. A.4.2–A.4.3 is assumption-free; when M fails, use the Holm fallback (validity preserved) and read the lower bound in compute form $C(\hat{\tau}) \geq \min_{\text{safe}} C_k$. A graphical procedure could be substituted in future work, but it is not evaluated here.

(iv) *Tighter p-values.* Replacing Hoeffding by Hoeffding–Bentkus or empirical-Bernstein shrinks $\epsilon_n(\delta)$, hence Δ_n , with no change to the argument.

APPENDIX B EXTENDED EXPERIMENTAL DETAILS

The controlled-simulation configuration of Section 10.1 uses a synthetic early-exit simulator with $L = 5$ exits and a $K = 40$ global-threshold chain. The E1/E3 validity study uses $n = 1000$ calibration examples and $T = 3000$ trials per (α, δ) cell. The E2/E7 cost comparison uses $n = 1000$ and $T = 2000$. The E5 joint-control study uses $n = 2000$, $T = 1000$, $\alpha = 0.0888475$, and $\delta = 0.10$. The E6 shift study uses $T = 400$, $\lambda = 2.0$, and a split of $n_1 = n_2 = 2500$.

The multi-chain redesign experiment E8 uses the same simulator with $L = 5$, six pre-specified per-exit threshold chains, 30 levels per chain, and 180 candidate policies. Oracle risk and cost are estimated from 250,000 simulator draws. The calibration study uses $n = 1000$, $T = 1000$, Hoeffding–Bentkus p-values, $\delta \in \{0.05, 0.10\}$, and targets $\alpha = R_{\min} + \{0.04, 0.08, 0.15\}$. The JSON artifact is `results/e8_multichain_selector.json`.

The digits tabular-cascade scalar-chain confirmation uses `sklearn.datasets.load_digits`, seed 20260615, a train/cache split with 899 cache examples, and a $K = 101$ scalar confidence-threshold chain. The exits are Gaussian naive Bayes, logistic regression, and support-vector classification, with relative costs $[1, 4, 12]$ and full-pool exit accuracies $[0.863, 0.968, 0.981]$. The fixed grid uses $\alpha \in \{0.05, 0.10, 0.15\}$, $\delta \in \{0.05, 0.10\}$, calibration sizes $\{100, 300\}$, Hoeffding–Bentkus p-values, $T = 1000$, and violation denominator “all”. The JSON artifact is `results/digits_tabular_real_cache_eval.json`.

The multi-chain real-cache reanalyses use the existing per-exit scores, losses, and costs stored in each cache, with six deterministic per-exit threshold schedules and 30 levels per chain. The digits multi-chain artifact is `results/digits_tabular_multichain_eval.json`. The breast-cancer and wine supporting artifacts are `results/breast_cancer_tabular_multichain_eval.json` and `results/wine_tabular_multichain_eval.json`; wine is retained as a saturated boundary case, not a main performance claim.

The CIFAR-10 feasible neural-cache benchmark uses the same Branchy ResNet-56-style adapter as the CIFAR-100 diagnostic, with exits after the stem, stage 1, stage 2, and stage 3. The model is trained for 30 epochs with seed 20260616 and evaluated on the CIFAR-10 test set. The cache contains $n = 10000$ examples, four exits, 101 scalar thresholds, top-softmax confidence, 0–1 loss, and cumulative analytical MAC estimates $[442,528, 42,909,856, 84,328,768, 125,747,840]$. Per-exit test accuracies are $[0.2551, 0.6928, 0.8750, 0.8825]$, and the full-cache multi-chain risk floor is about 0.1173. The multi-chain evaluation uses six deterministic per-exit threshold schedules, 30 levels per chain, $K = 180$ policies, $\alpha \in \{0.12, 0.15, 0.20, 0.25, 0.30\}$, $\delta \in \{0.05, 0.10\}$, calibration sizes $\{500, 2000\}$, Hoeffding–Bentkus p-values, $T = 1000$, and violation denominator “all”. The JSON artifact is `results/cifar10_branchy_multichain_eval.json`.

The CIFAR-100 diagnostic uses a Branchy ResNet-56-style model with exits after the stem, stage 1, stage 2, and stage 3. The cache contains $n = 10000$ test examples, $K = 100$ scalar thresholds, top-softmax confidence, 0–1 loss, and cumulative analytical MAC estimates $[443,968, 42,911,296, 84,331,648, 125,753,600]$. The checkpoint records epoch 80 and per-exit test accuracies $[0.0646, 0.3040, 0.6147, 0.6945]$. The full-pool chain cost is nondecreasing; the full-pool chain risk has four adjacent increases, maximum adjacent increase 2×10^{-4} , and minimum risk 0.3055. The CIFAR multi-chain reanalysis uses the same cache with 180 candidate policies; its JSON artifact is `results/cifar_branchy_multichain_eval.json`.

A. EMBEDDED NUMERICAL RESULTS

Tables 5–10 embed the synthetic and CIFAR numerical outputs used in Section 10; Table 4 summarizes the digits confirmation. The submitted manuscript therefore does not rely on local result files for the reported headline numbers. A dash indicates that a method did not select a policy in that cell, so conditional mean risk or cost is not applicable. These appendices are typeset single-column because the proof displays and the wide, multi-column row-level tables do not fit the body text width; the headline numbers they support are reproduced in the main-text Table 4 and Figure 3, so the two-column body is self-contained.

TABLE 5: Synthetic E1/E3 validity and naive-baseline results. Violation entries are counts/rates over all $T = 3000$ trials.

α	δ	Chain feas.	Chain viol.	Chain CI	Chain cost	Naive feas.	Naive viol.	Naive cost
0.079	0.05	2578/3000 (0.859)	3/3000 (0.001)	[0.000,0.003]	3.509	3000/3000 (1.000)	694/3000 (0.231)	3.383
0.079	0.10	2734/3000 (0.911)	5/3000 (0.002)	[0.001,0.004]	3.492	3000/3000 (1.000)	694/3000 (0.231)	3.383
0.119	0.05	3000/3000 (1.000)	18/3000 (0.006)	[0.004,0.009]	3.276	3000/3000 (1.000)	1054/3000 (0.351)	3.183
0.119	0.10	3000/3000 (1.000)	43/3000 (0.014)	[0.010,0.019]	3.262	3000/3000 (1.000)	1054/3000 (0.351)	3.183
0.189	0.05	3000/3000 (1.000)	16/3000 (0.005)	[0.003,0.009]	3.024	3000/3000 (1.000)	912/3000 (0.304)	2.944

α	δ	Chain feas.	Chain viol.	Chain CI	Chain cost	Naive feas.	Naive viol.	Naive cost
0.189	0.10	3000/3000 (1.000)	35/3000 (0.012)	[0.008,0.016]	3.011	3000/3000 (1.000)	912/3000 (0.304)	2.944

TABLE 6: Synthetic E2/E7 chain, Holm, Bonferroni, and p-value ablation. Here $\alpha = 0.1188475$, $\delta = 0.10$, $n = 1000$, and $T = 2000$.

Method	p-value	Feas.	Viol.	CI	Mean cost	n	T	α
chain	hb	2000/2000 (1.000)	32/2000 (0.016)	[0.011,0.023]	3.260	1000	2000	0.119
chain	hoeffding	2000/2000 (1.000)	0/2000 (0.000)	[0.000,0.002]	3.344	1000	2000	0.119
holm	hb	2000/2000 (1.000)	1/2000 (0.001)	[0.000,0.003]	3.311	1000	2000	0.119
holm	hoeffding	2000/2000 (1.000)	0/2000 (0.000)	[0.000,0.002]	3.459	1000	2000	0.119
bonferroni	hb	2000/2000 (1.000)	1/2000 (0.001)	[0.000,0.003]	3.326	1000	2000	0.119
bonferroni	hoeffding	2000/2000 (1.000)	0/2000 (0.000)	[0.000,0.002]	3.487	1000	2000	0.119

TABLE 7: Synthetic E5 joint risk–compute control. The oracle cheapest safe cost is 3.312705; $\alpha = 0.0888475$, $\delta = 0.10$, $n = 2000$, and $T = 1000$.

Budget case	B	Oracle feasible	Feas.	Risk viol.	Risk CI	Mean UCB	Mean true cost
slack	4.641	yes	1000/1000 (1.000)	0/1000 (0.000)	[0.000,0.004]	3.620	3.387
tight_oracle_feasible	3.363	yes	0/1000 (0.000)	0/1000 (0.000)	[0.000,0.004]	–	–

TABLE 8: Synthetic E6 shift results. Weighted-arm violation rates are conditional on feasible selections; when feasibility is zero, the rate is not applicable.

Arm	α	η	$\ \hat{w} - w\ _1$ (*)	Feas.	Viol.	CI	δ	Note
unweighted	0.161	–	–	no	400/400 (1.000)	350/400 (0.875)	[0.839,0.906]	0.10 source-calibrated
estimated	–	0.00	0.060	no	400/400 (1.000)	0/400 (0.000)	[0.000,0.009]	– eta sweep
exact	–	0.00	0.000	yes	400/400 (1.000)	0/400 (0.000)	[0.000,0.009]	– oracle weights
estimated	–	0.03	0.060	no	400/400 (1.000)	0/400 (0.000)	[0.000,0.009]	– eta sweep
exact	–	0.03	0.000	yes	400/400 (1.000)	0/400 (0.000)	[0.000,0.009]	– oracle weights
estimated	–	0.06	0.060	no	140/400 (0.350)	0/140 (0.000)	[0.000,0.026]	– eta sweep
exact	–	0.06	0.000	yes	400/400 (1.000)	0/400 (0.000)	[0.000,0.009]	– oracle weights
estimated	–	0.09	0.060	yes	0/400 (0.000)	0/0 (–)	[0.000,0.975]	– eta sweep
exact	–	0.09	0.000	yes	400/400 (1.000)	0/400 (0.000)	[0.000,0.009]	– oracle weights
estimated	–	0.12	0.060	yes	0/400 (0.000)	0/0 (–)	[0.000,0.975]	– eta sweep
exact	–	0.12	0.000	yes	400/400 (1.000)	0/400 (0.000)	[0.000,0.009]	– oracle weights
biased	0.091	0.02	0.351	no	400/400 (1.000)	358/400 (0.895)	[0.861,0.923]	0.10 failure mode

TABLE 9: CIFAR-100 fixed-grid finite-pool cache diagnostic. The run uses HB p-values, $T = 1000$, violation denominator “all”, scalar top-softmax thresholds on a $K = 100$ chain, and analytical MAC estimates. None of the pre-defined targets has a truly safe full-pool chain configuration because the minimum full-pool chain risk is 0.3055. Cells with $\alpha \in \{0.05, 0.10, 0.20\}$ are condensed into a single line because every such row is identically zero; the informative $\alpha = 0.30$ cells are listed in full.

α	δ	n_{cal}	n_{test}	Method	Feas.	Viol.	CI	Risk	Cost (M MACs)
<i>For every cell with $\alpha \in \{0.05, 0.10, 0.20\}$—all four methods, both $n_{cal} \in \{500, 2000\}$, and both $\delta \in \{0.05, 0.10\}$ (48 rows)—feasibility and violation are 0/1000 (CI [0.000, 0.004]) and no policy is selected, since the minimum full-pool chain risk 0.3055 exceeds each such α: no configuration certifies, and naive tuning never finds an empirically safe policy. The informative $\alpha = 0.30$ cells are listed in full below.</i>									
0.30	0.05	500	9500	chain	4/1000 (0.004)	4/1000 (0.004)	[0.001,0.010]	0.311	112.11
0.30	0.05	500	9500	holm	0/1000 (0.000)	0/1000 (0.000)	[0.000,0.004]	–	–
0.30	0.05	500	9500	bonferroni	0/1000 (0.000)	0/1000 (0.000)	[0.000,0.004]	–	–
0.30	0.05	500	9500	naive	457/1000 (0.457)	457/1000 (0.457)	[0.426,0.488]	0.322	104.63
0.30	0.05	2000	8000	chain	0/1000 (0.000)	0/1000 (0.000)	[0.000,0.004]	–	–
0.30	0.05	2000	8000	holm	0/1000 (0.000)	0/1000 (0.000)	[0.000,0.004]	–	–
0.30	0.05	2000	8000	bonferroni	0/1000 (0.000)	0/1000 (0.000)	[0.000,0.004]	–	–
0.30	0.05	2000	8000	naive	281/1000 (0.281)	281/1000 (0.281)	[0.253,0.310]	0.314	110.36
0.30	0.10	500	9500	chain	13/1000 (0.013)	13/1000 (0.013)	[0.007,0.022]	0.315	110.99
0.30	0.10	500	9500	holm	0/1000 (0.000)	0/1000 (0.000)	[0.000,0.004]	–	–
0.30	0.10	500	9500	bonferroni	0/1000 (0.000)	0/1000 (0.000)	[0.000,0.004]	–	–
0.30	0.10	500	9500	naive	434/1000 (0.434)	434/1000 (0.434)	[0.403,0.465]	0.322	104.55
0.30	0.10	2000	8000	chain	4/1000 (0.004)	4/1000 (0.004)	[0.001,0.010]	0.316	112.53
0.30	0.10	2000	8000	holm	0/1000 (0.000)	0/1000 (0.000)	[0.000,0.004]	–	–
0.30	0.10	2000	8000	bonferroni	0/1000 (0.000)	0/1000 (0.000)	[0.000,0.004]	–	–
0.30	0.10	2000	8000	naive	302/1000 (0.302)	302/1000 (0.302)	[0.274,0.332]	0.314	110.88

TABLE 10: CIFAR-100 exploratory feasible-target finite-pool cache diagnostic. This post hoc run uses HB p-values, $T = 1000$, violation denominator “all”, the same cache and chain as Table 9, and targets $\alpha \in \{0.35, 0.40\}$.

α	δ	n_{cal}	n_{test}	Method	Feas.	Viol.	CI	Risk	Cost (M MACs)
0.35	0.05	500	9500	chain	484/1000 (0.484)	17/1000 (0.017)	[0.010,0.027]	0.321	105.30
0.35	0.05	500	9500	holm	75/1000 (0.075)	1/1000 (0.001)	[0.000,0.006]	0.318	106.79
0.35	0.05	500	9500	bonferroni	75/1000 (0.075)	1/1000 (0.001)	[0.000,0.006]	0.317	107.42
0.35	0.05	500	9500	naive	990/1000 (0.990)	472/1000 (0.472)	[0.441,0.503]	0.350	92.85
0.35	0.05	2000	8000	chain	996/1000 (0.996)	20/1000 (0.020)	[0.012,0.031]	0.327	100.62
0.35	0.05	2000	8000	holm	789/1000 (0.789)	1/1000 (0.001)	[0.000,0.006]	0.317	106.52
0.35	0.05	2000	8000	bonferroni	789/1000 (0.789)	0/1000 (0.000)	[0.000,0.004]	0.316	107.01
0.35	0.05	2000	8000	naive	1000/1000 (1.000)	441/1000 (0.441)	[0.410,0.472]	0.349	92.42
0.35	0.10	500	9500	chain	616/1000 (0.616)	38/1000 (0.038)	[0.027,0.052]	0.324	103.34
0.35	0.10	500	9500	holm	110/1000 (0.110)	2/1000 (0.002)	[0.000,0.007]	0.318	107.33
0.35	0.10	500	9500	bonferroni	110/1000 (0.110)	1/1000 (0.001)	[0.000,0.006]	0.317	107.81
0.35	0.10	500	9500	naive	989/1000 (0.989)	506/1000 (0.506)	[0.475,0.537]	0.352	92.42
0.35	0.10	2000	8000	chain	995/1000 (0.995)	37/1000 (0.037)	[0.026,0.051]	0.330	99.35
0.35	0.10	2000	8000	holm	835/1000 (0.835)	4/1000 (0.004)	[0.001,0.010]	0.318	105.87
0.35	0.10	2000	8000	bonferroni	835/1000 (0.835)	1/1000 (0.001)	[0.000,0.006]	0.317	106.41
0.35	0.10	2000	8000	naive	1000/1000 (1.000)	453/1000 (0.453)	[0.422,0.484]	0.349	92.51
0.40	0.05	500	9500	chain	994/1000 (0.994)	12/1000 (0.012)	[0.006,0.021]	0.353	92.25
0.40	0.05	500	9500	holm	824/1000 (0.824)	0/1000 (0.000)	[0.000,0.004]	0.333	99.13
0.40	0.05	500	9500	bonferroni	824/1000 (0.824)	0/1000 (0.000)	[0.000,0.004]	0.329	100.91
0.40	0.05	500	9500	naive	1000/1000 (1.000)	529/1000 (0.529)	[0.498,0.560]	0.399	81.28
0.40	0.05	2000	8000	chain	1000/1000 (1.000)	23/1000 (0.023)	[0.015,0.034]	0.375	86.35
0.40	0.05	2000	8000	holm	1000/1000 (1.000)	2/1000 (0.002)	[0.000,0.007]	0.362	89.15
0.40	0.05	2000	8000	bonferroni	1000/1000 (1.000)	0/1000 (0.000)	[0.000,0.004]	0.359	89.79
0.40	0.05	2000	8000	naive	1000/1000 (1.000)	462/1000 (0.462)	[0.431,0.493]	0.398	81.52
0.40	0.10	500	9500	chain	995/1000 (0.995)	27/1000 (0.027)	[0.018,0.039]	0.358	90.59
0.40	0.10	500	9500	holm	867/1000 (0.867)	0/1000 (0.000)	[0.000,0.004]	0.335	98.29
0.40	0.10	500	9500	bonferroni	867/1000 (0.867)	0/1000 (0.000)	[0.000,0.004]	0.331	99.91
0.40	0.10	500	9500	naive	1000/1000 (1.000)	505/1000 (0.505)	[0.474,0.536]	0.399	81.32
0.40	0.10	2000	8000	chain	1000/1000 (1.000)	16/1000 (0.016)	[0.009,0.026]	0.378	85.65
0.40	0.10	2000	8000	holm	1000/1000 (1.000)	0/1000 (0.000)	[0.000,0.004]	0.364	88.68
0.40	0.10	2000	8000	bonferroni	1000/1000 (1.000)	0/1000 (0.000)	[0.000,0.004]	0.361	89.33
0.40	0.10	2000	8000	naive	1000/1000 (1.000)	445/1000 (0.445)	[0.414,0.476]	0.398	81.53

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